

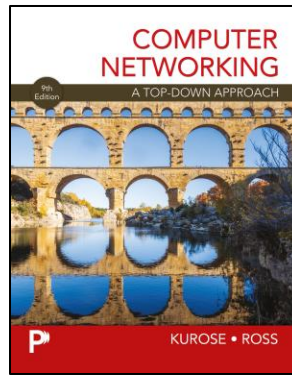
CSC 361

Computer Networks

Overview

Wenjun Yang

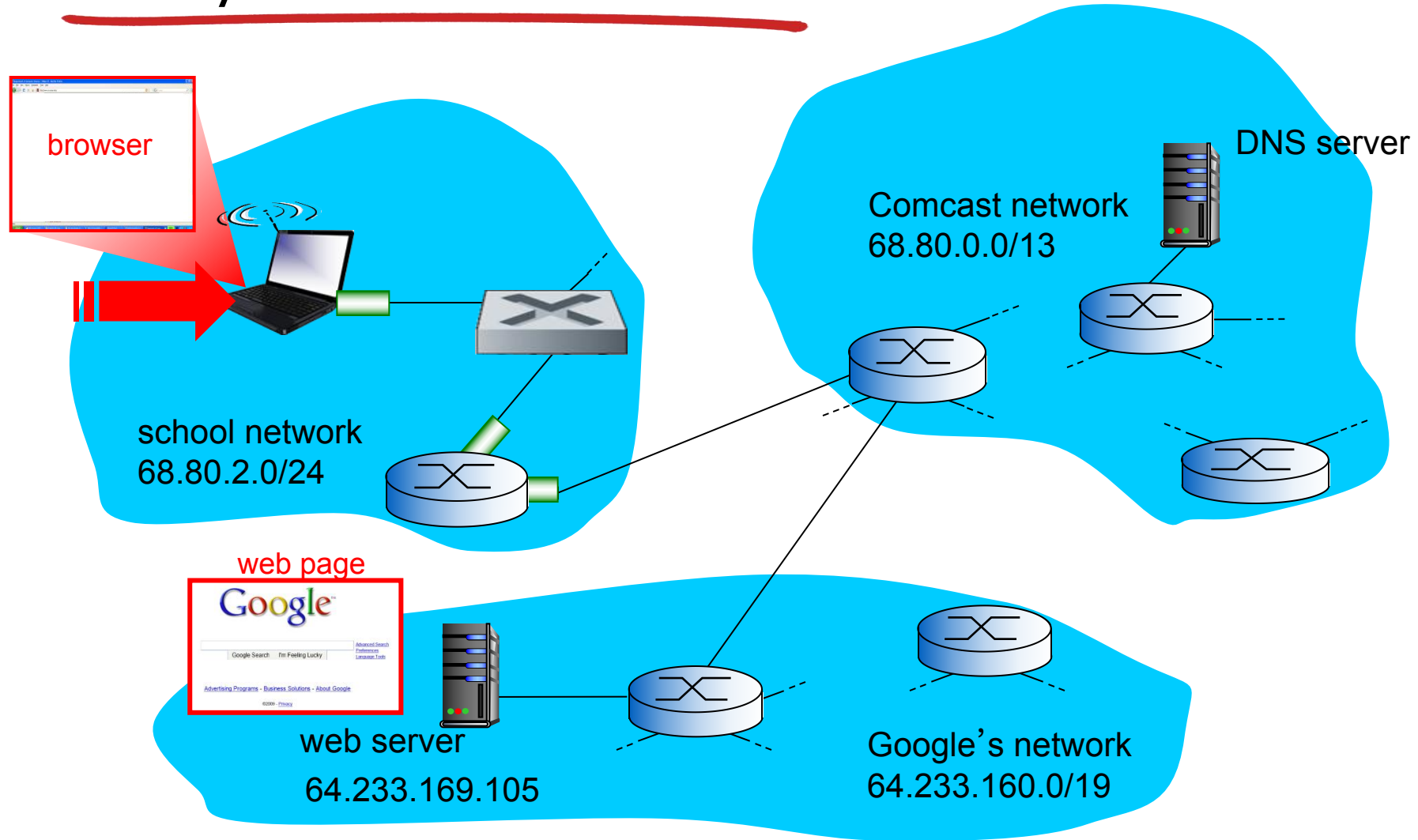
Summer 2026



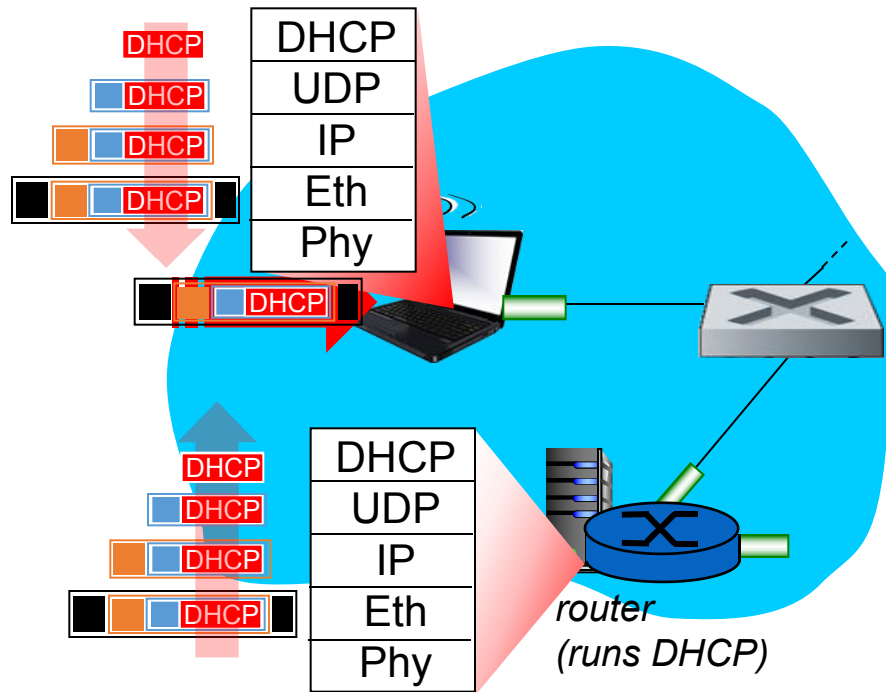
Synthesis: a day in the life of a web request

- journey down protocol stack complete!
 - application, transport, network, link
- putting-it-all-together: synthesis!
 - *goal:* identify, review, understand protocols (at all layers) involved in seemingly simple scenario: requesting www page
 - *scenario:* student attaches laptop to campus network, requests/receives www.google.com

A day in the life: scenario

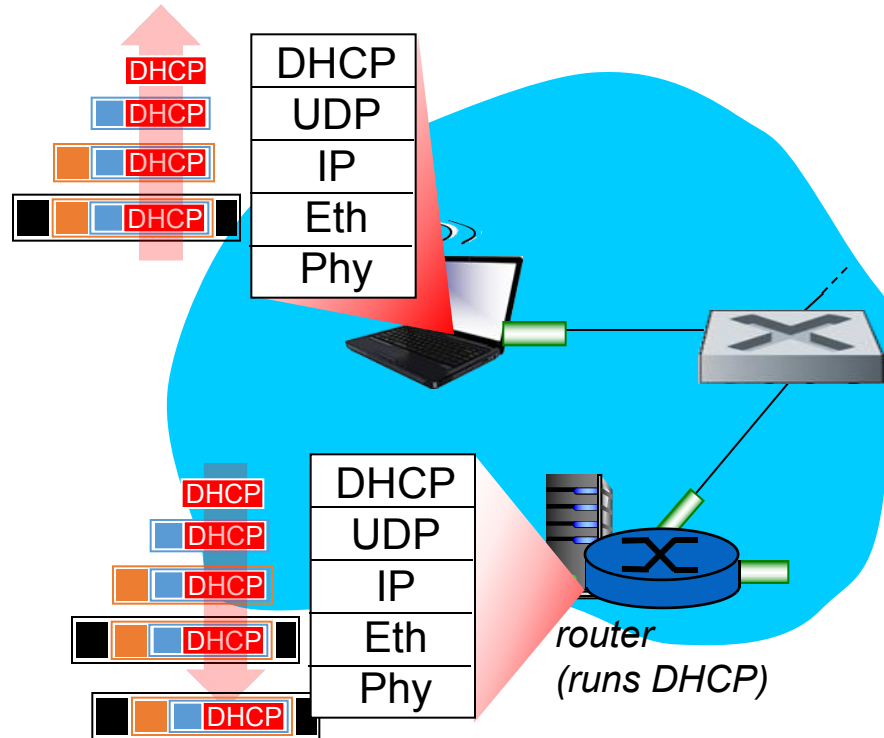


A day in the life... connecting to the Internet



- connecting laptop needs to get its own IP address, addr of first-hop router, addr of DNS server: use *DHCP*
- DHCP request encapsulated in *UDP*, encapsulated in *IP*, encapsulated in *802.3 Ethernet*
- Ethernet frame *broadcast* (dest: FFFFFFFFFFFFFFFF) on LAN, received at router running *DHCP* server
- Ethernet *demuxed* to IP demuxed, UDP demuxed to DHCP

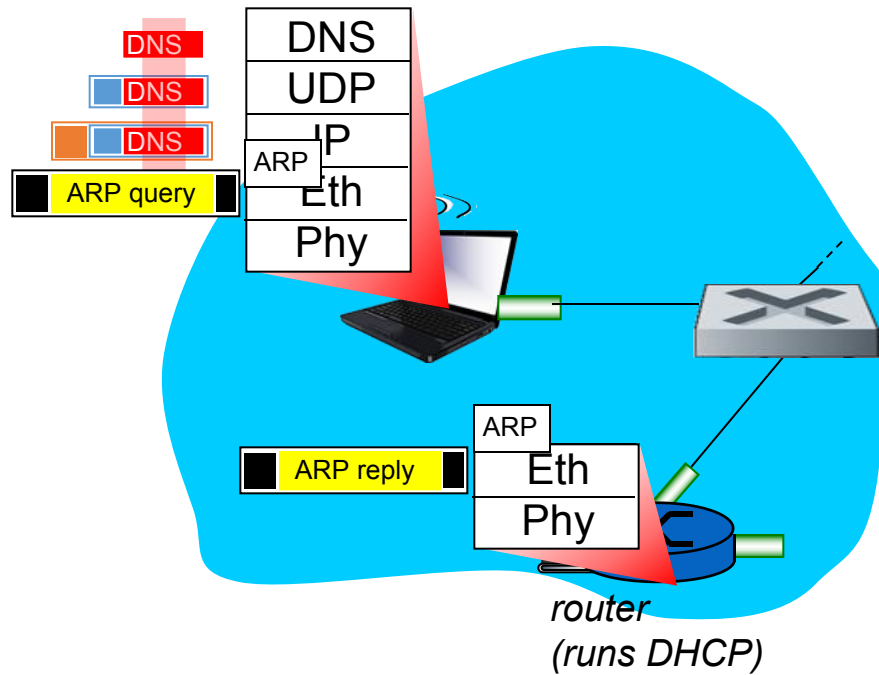
A day in the life... connecting to the Internet



- DHCP server formulates **DHCP ACK** containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulation at DHCP server, frame forwarded (**switch learning**) through LAN, demultiplexing at client
- DHCP client receives DHCP ACK reply

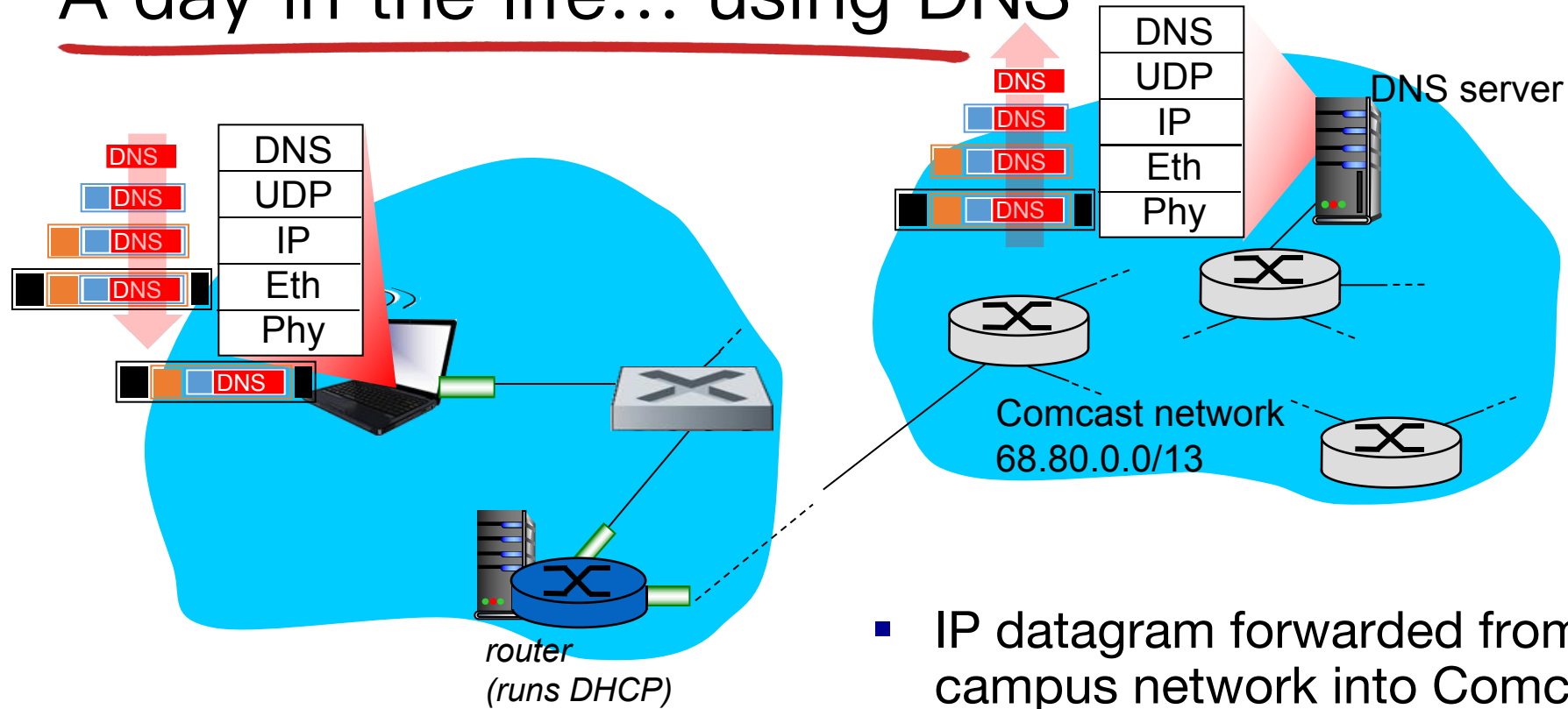
Client now has IP address, knows name & addr of DNS server, IP address of its first-hop router

A day in the life... ARP (before DNS, before HTTP)



- before sending *HTTP* request, need IP address of `www.google.com`: *DNS*
- DNS query created, encapsulated in UDP, encapsulated in IP, encapsulated in Eth. To send frame to router, need MAC address of router interface: *ARP*
- *ARP query* broadcast, received by router, which replies with *ARP reply* giving MAC address of router interface
- client now knows MAC address of first hop router, so can now send frame containing DNS query

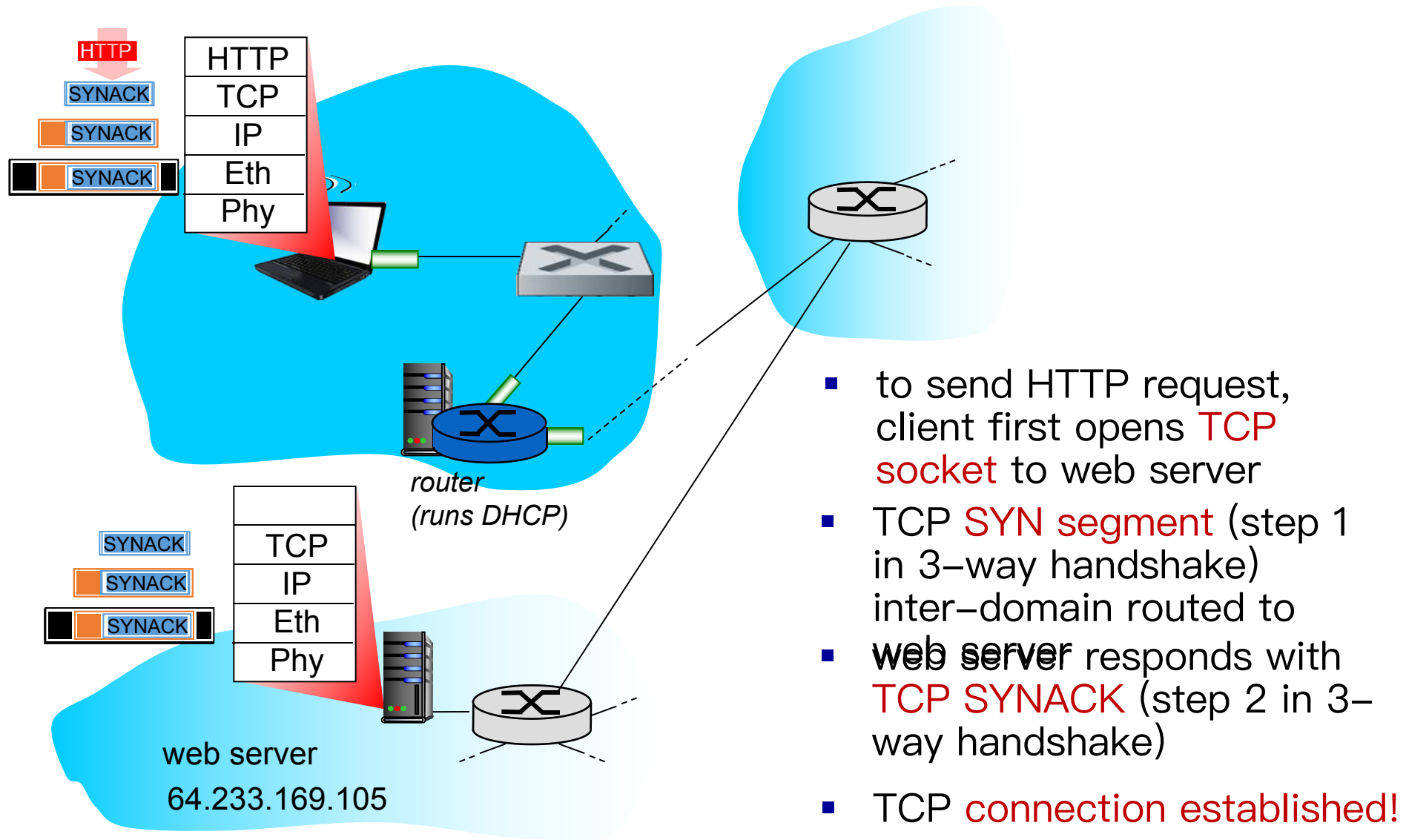
A day in the life... using DNS



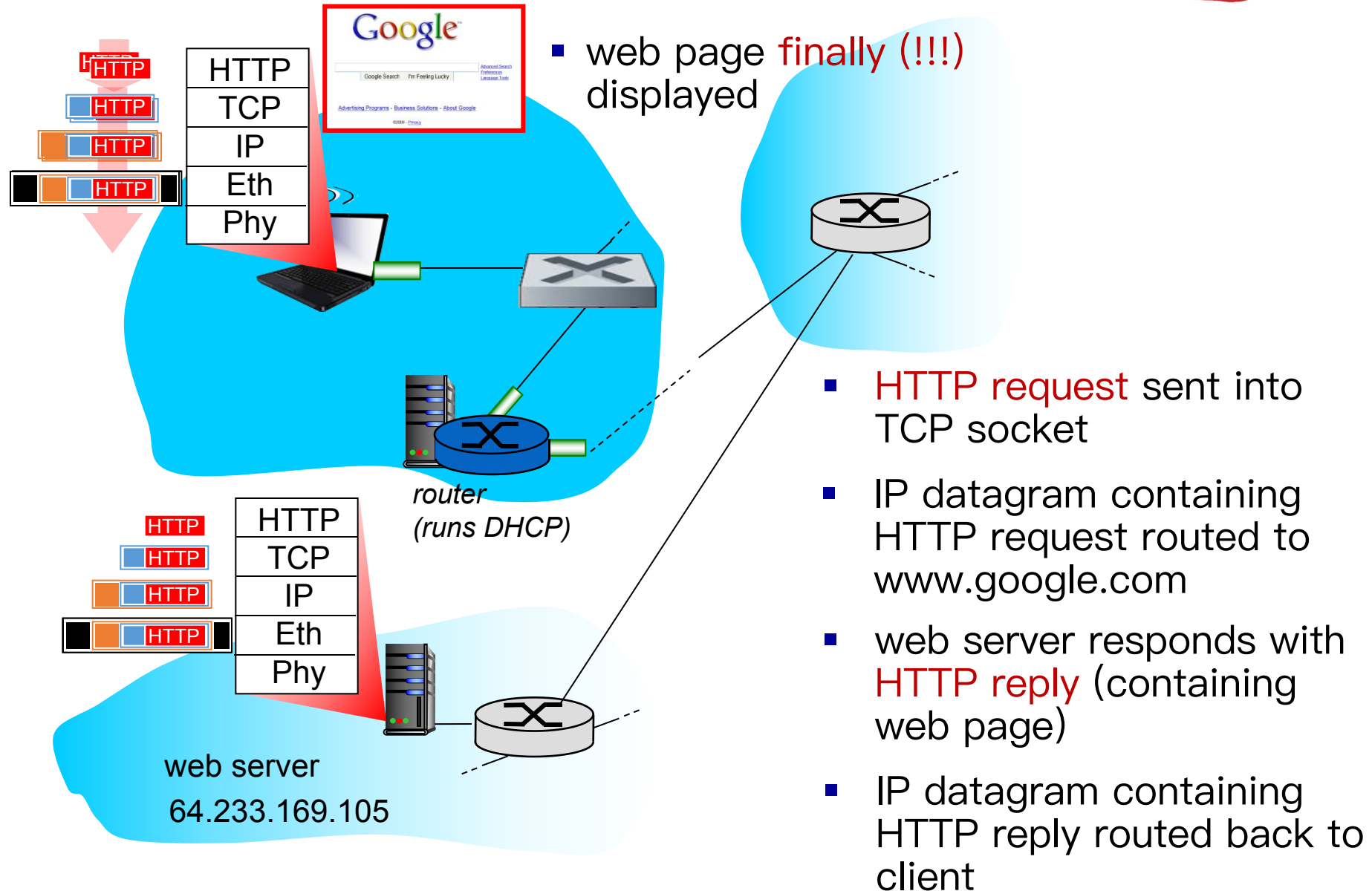
- IP datagram containing DNS query forwarded via LAN switch from client to 1st hop router

- IP datagram forwarded from campus network into Comcast network, routed (tables created by **RIP**, **OSPF**, **IS-IS** and/or **BGP** routing protocols) to DNS server
- demuxed to DNS server
- DNS server replies to client with IP address of www.google.com

A day in the life...TCP connection carrying HTTP



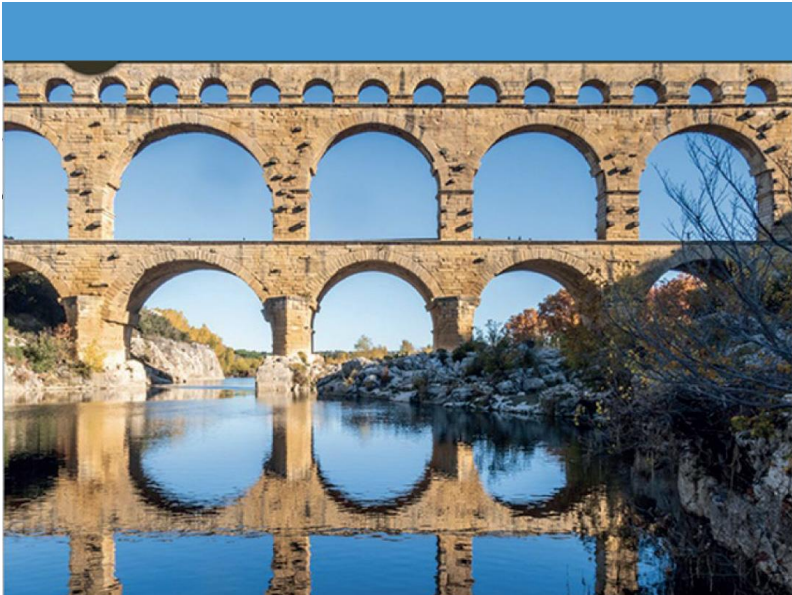
A day in the life... HTTP request/reply



Chapter 1: Introduction

Chapter goal:

- Get “feel,” “big picture,” introduction to terminology
 - more depth, detail **later** in course



Overview/roadmap:

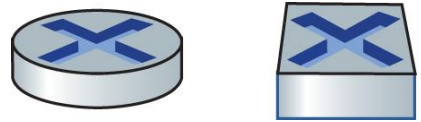
- What is the Internet? What is a protocol?
- **Network edge:** hosts, access network, physical media
- **Network core:** packet/circuit switching, internet structure
- **Performance:** loss, delay, throughput
- Protocol layers, service models

The Internet: a “Nuts and Bolts” View (1 of 2)



Billions of connected computing **devices**:

- **hosts** = end systems
- running **network apps** at Internet's “edge”



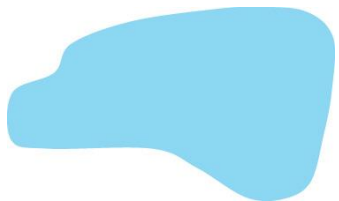
Packet switches: forward packets (chunks of data)

- routers, switches



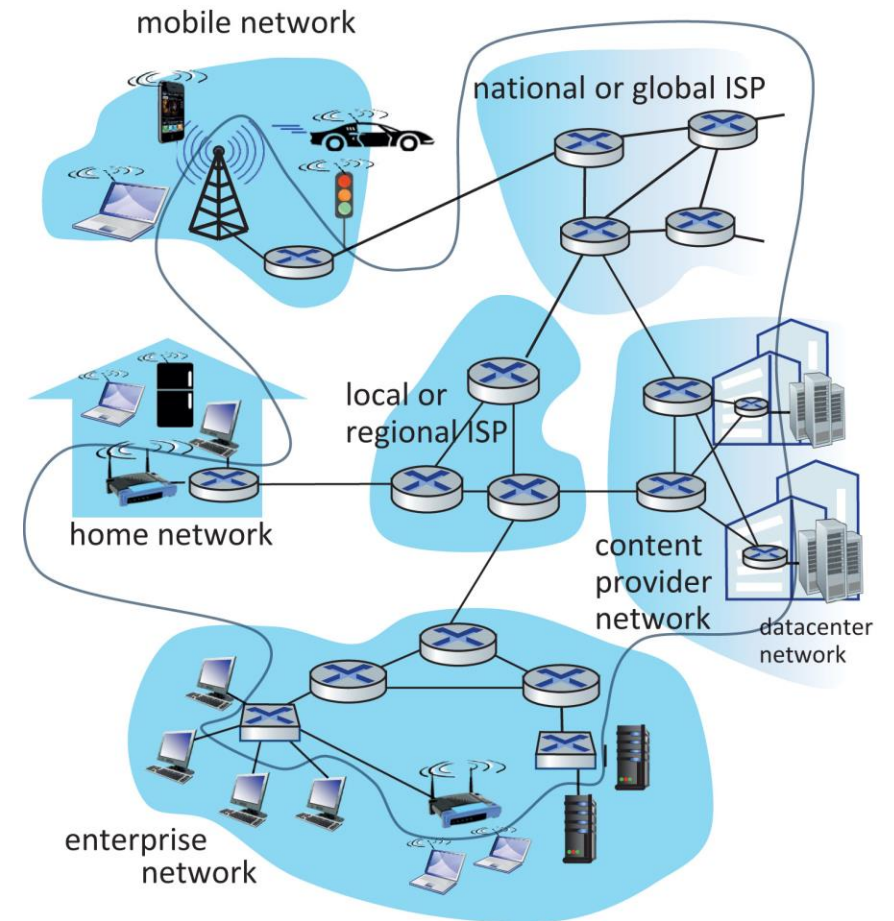
Communication links

- fiber, copper, radio, satellite
- transmission rate: **bandwidth**

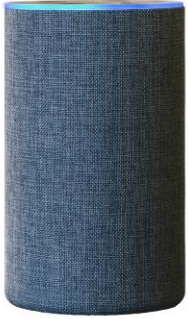


Networks

- collection of devices, routers, links: managed by an organization



“Fun” Internet-Connected Devices (1 of 2)



Amazon Echo



Security Camera



Internet phones



Internet refrigerator



Gaming devices



IP picture frame



Slingbox: remote control cable TV



Pacemaker & Monitor



Web-enabled toaster + Weather forecaster

“Fun” Internet-Connected Devices (2 of 2)



sensorized, bed
mattress



Fitbit



AR devices



diapers



Tweet-a-watt:
monitor
energy use



bikes



cars

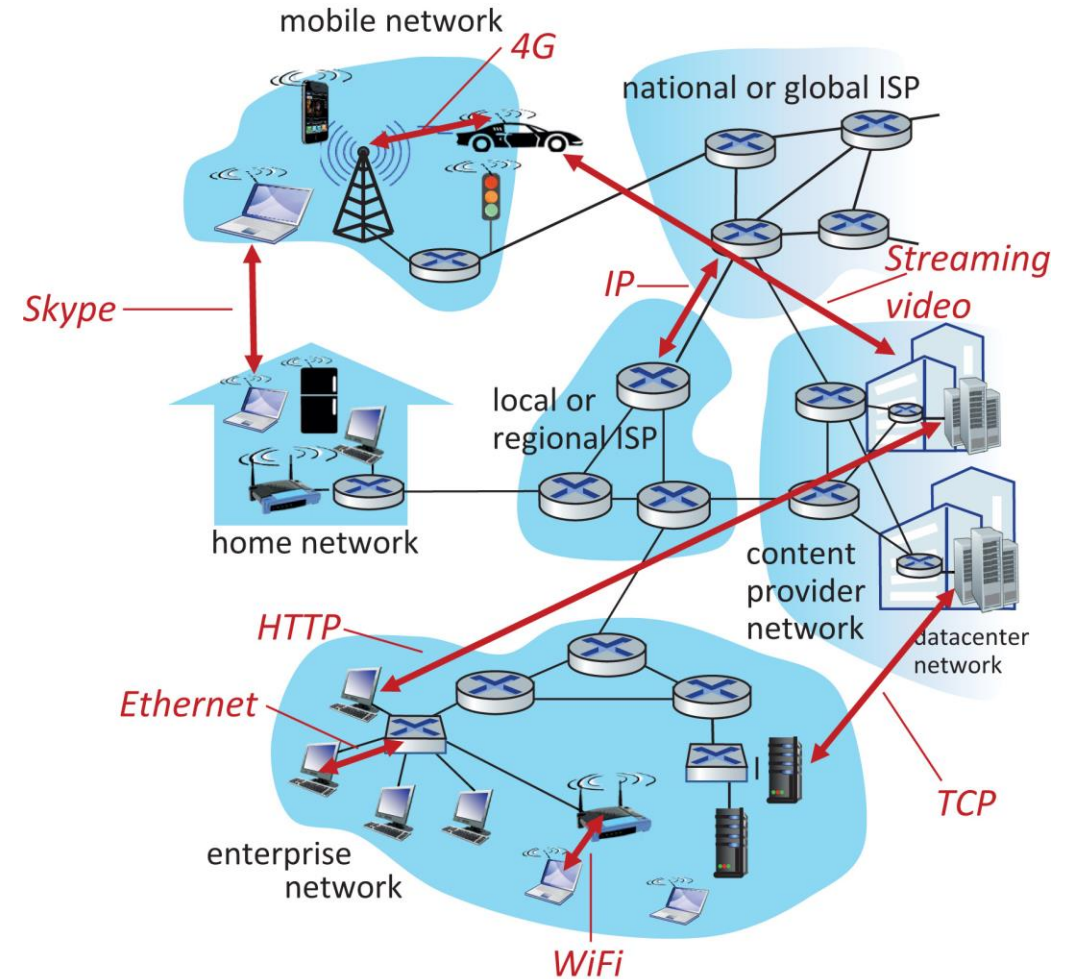


scooters

Others?

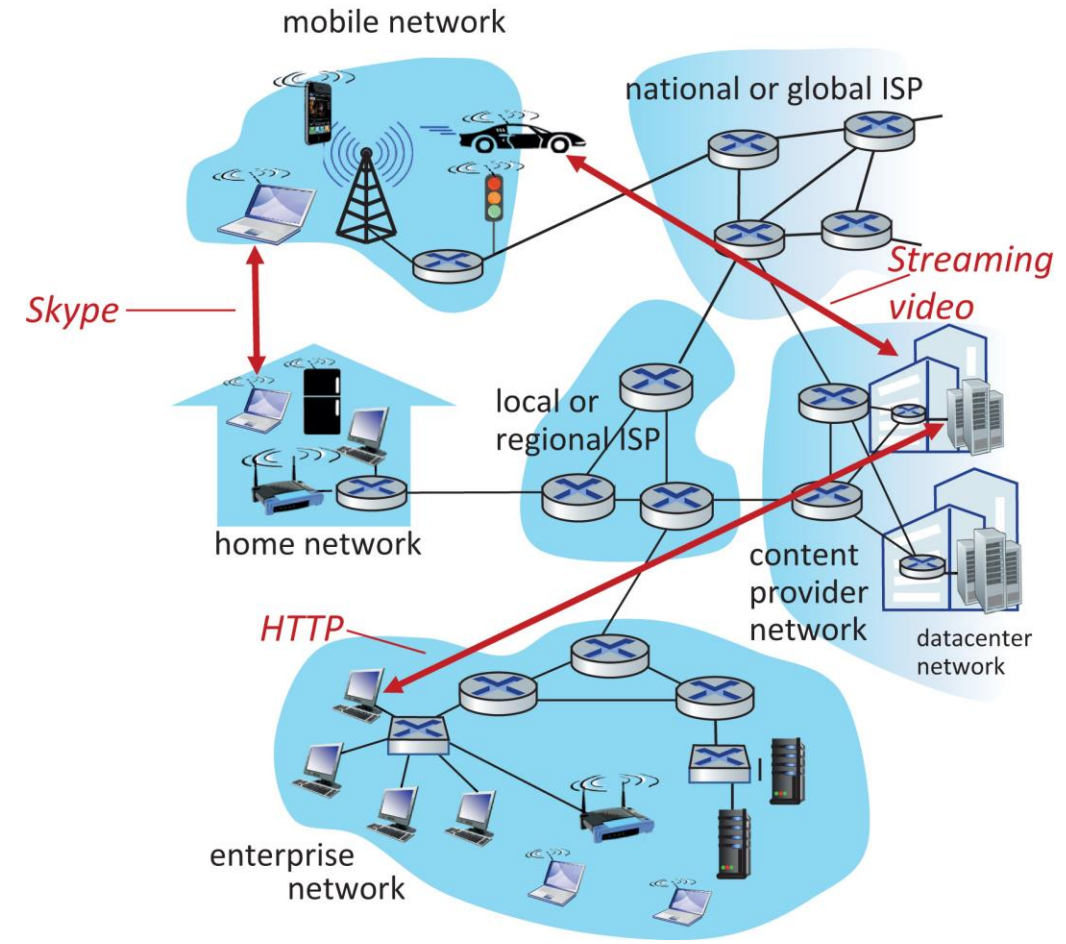
The Internet: a “Nuts and Bolts” View (2 of 2)

- **Internet: “network of networks”**
 - Interconnected ISPs
- **protocols** are **everywhere**
 - control sending, receiving of messages
 - e.g., HTTP (Web), streaming video, Zoom, TCP, IP, WiFi, 4/5G, Ethernet
- **Internet standards**
 - RFC: Request for Comments
 - IETF: Internet Engineering Task Force



The Internet: a “Services” View

- **Infrastructure** that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides **programming interface** to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service



What's a Protocol? (1 of 2)

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

Rules for:

- ... specific messages sent
- ... specific actions taken when message received, or other events

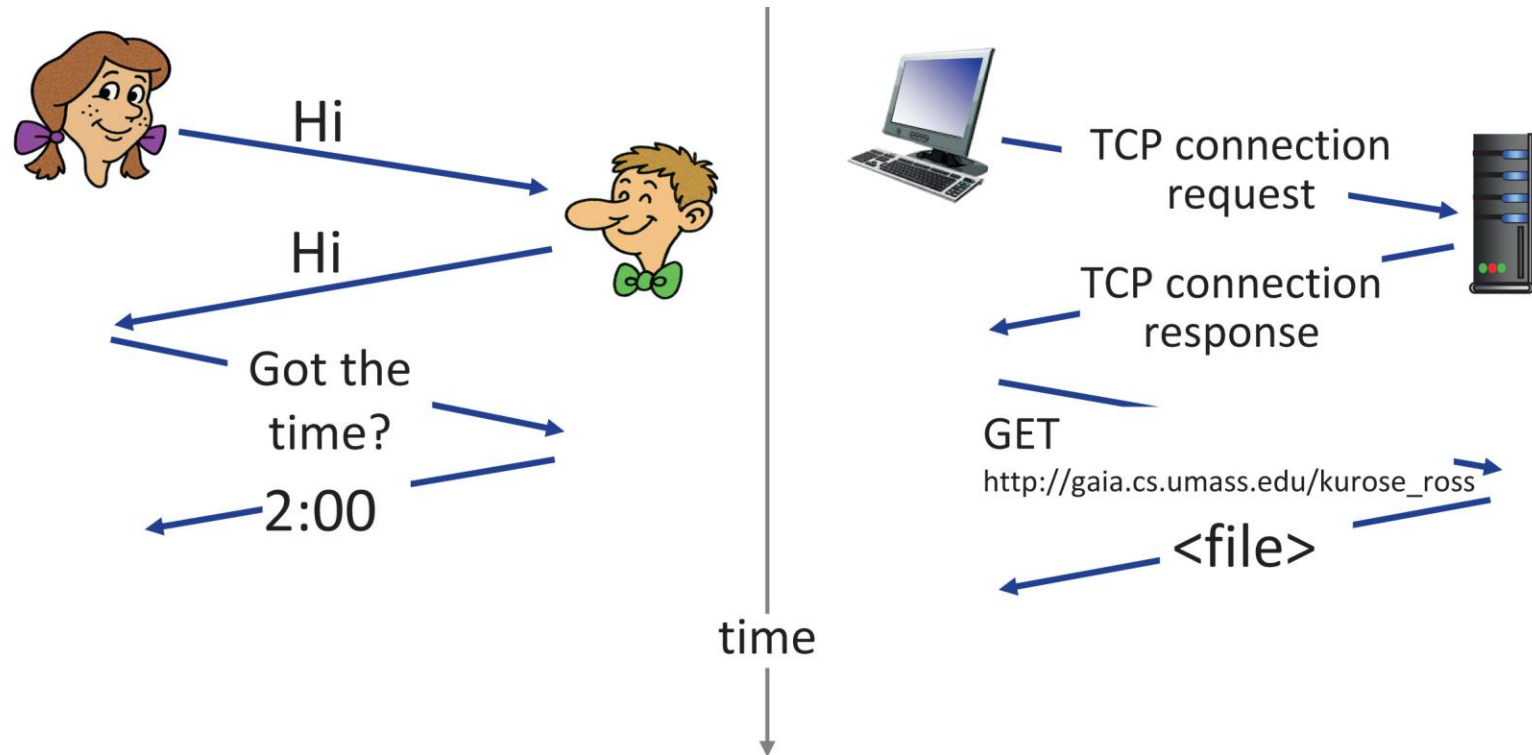
Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

Protocols define the **format, order of messages sent and received** among network entities, and **actions taken** on message transmission, receipt

What's a Protocol? (2 of 2)

A human protocol and a computer network protocol:



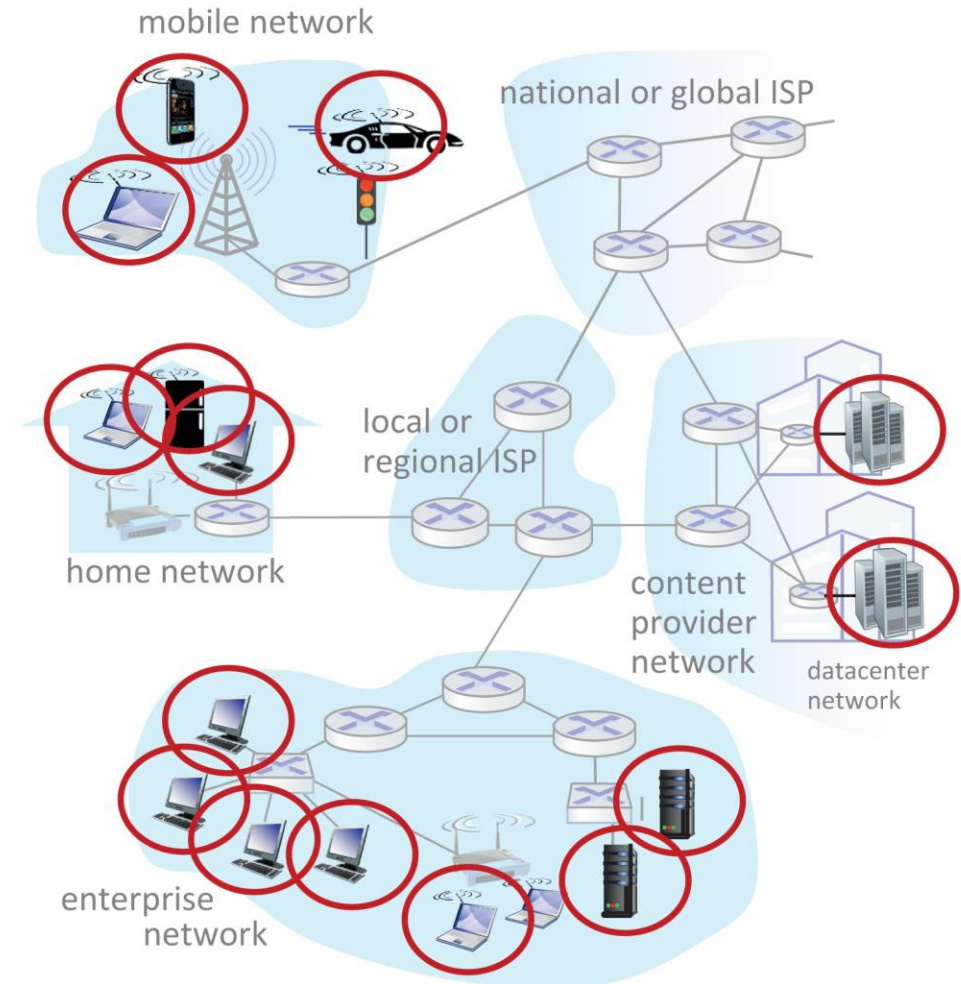
Chapter 1: Roadmap (1 of 6)

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- History

A Closer look at Internet Structure (1 of 3)

Network edge:

- hosts: clients and servers
- servers often in data centers



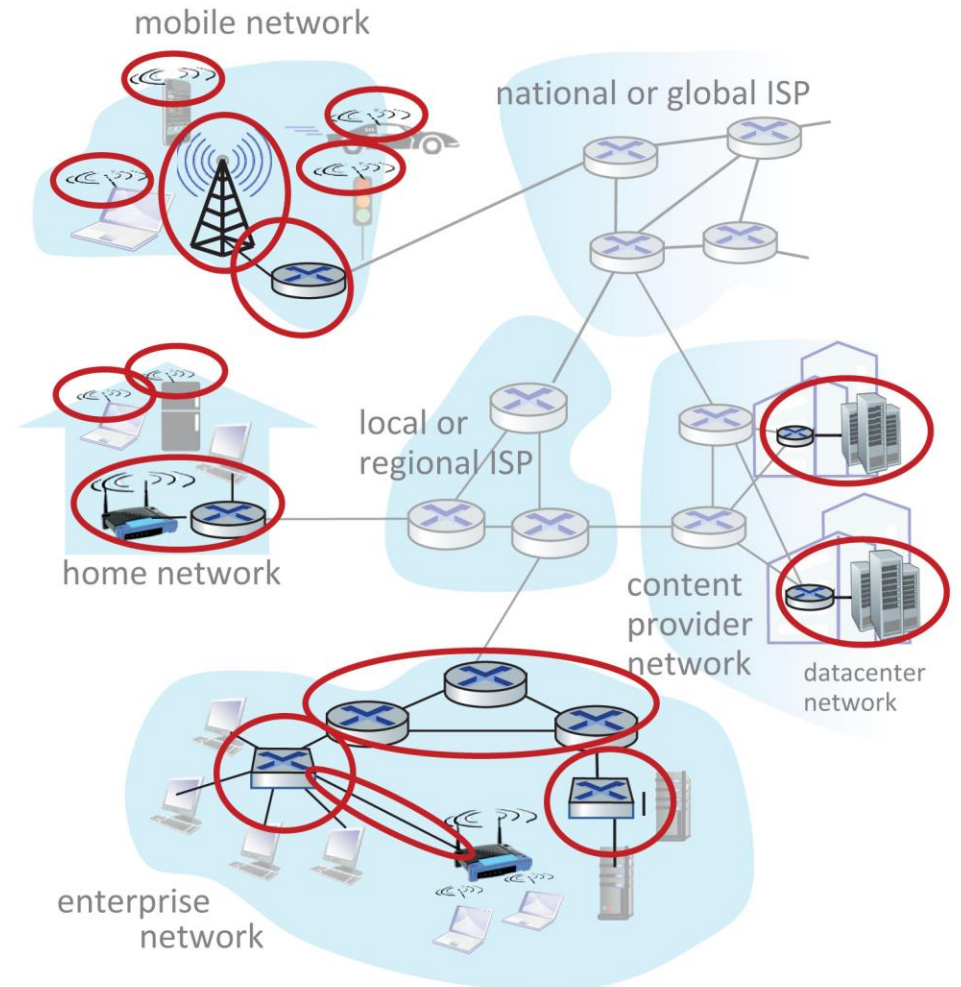
A Closer look at Internet Structure (2 of 3)

Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links



A Closer look at Internet Structure (3 of 3)

Network edge:

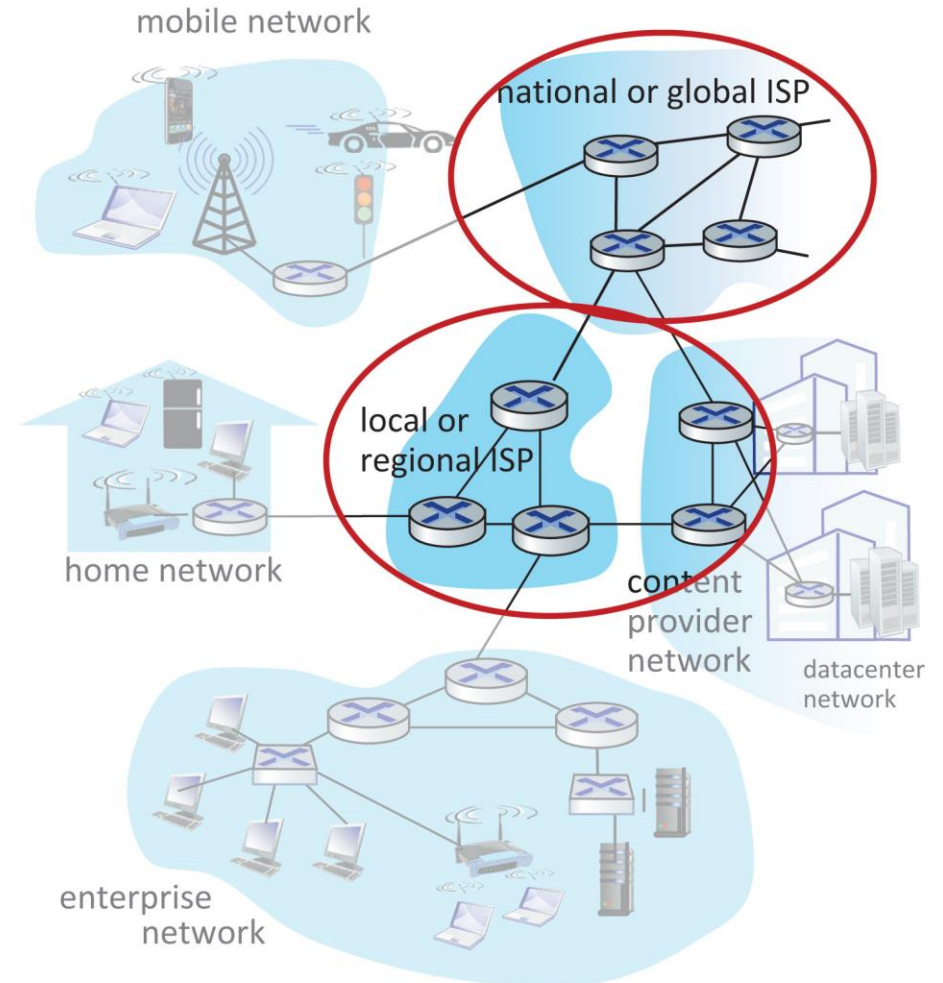
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Access networks, physical media:

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Network core:

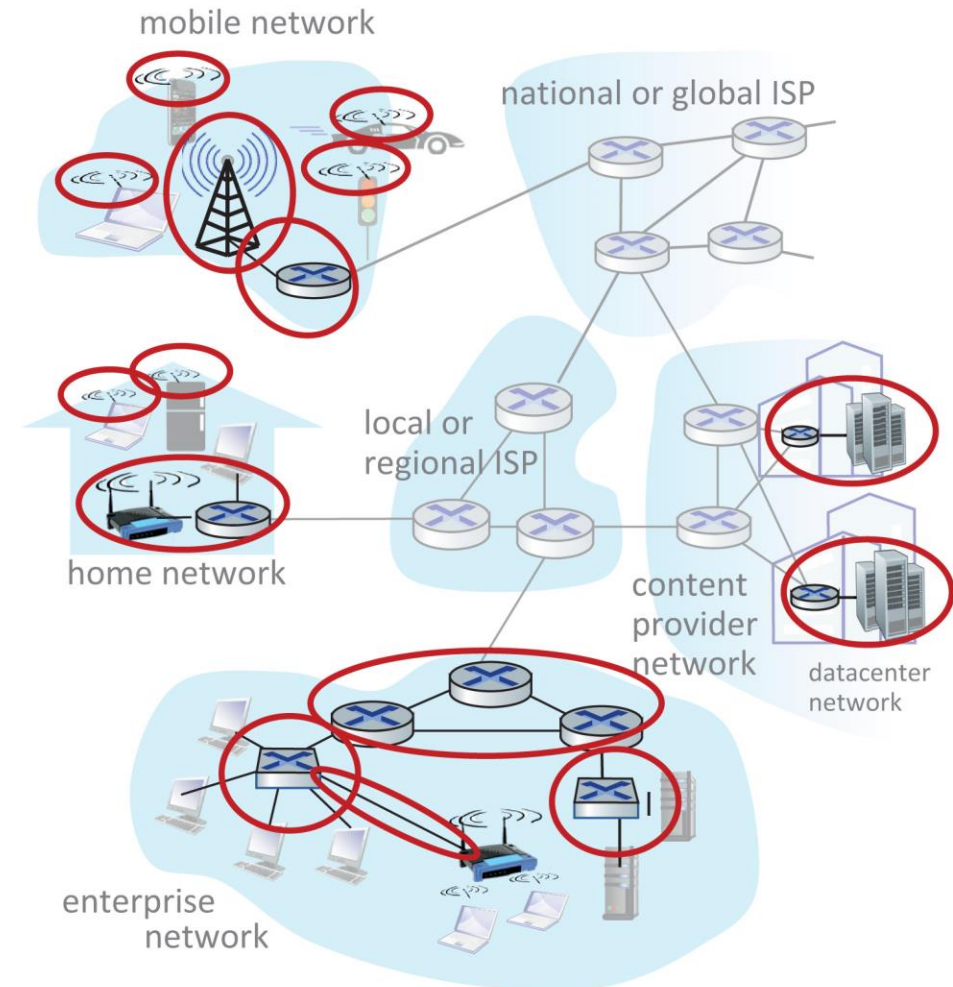
- interconnected routers
- network of networks



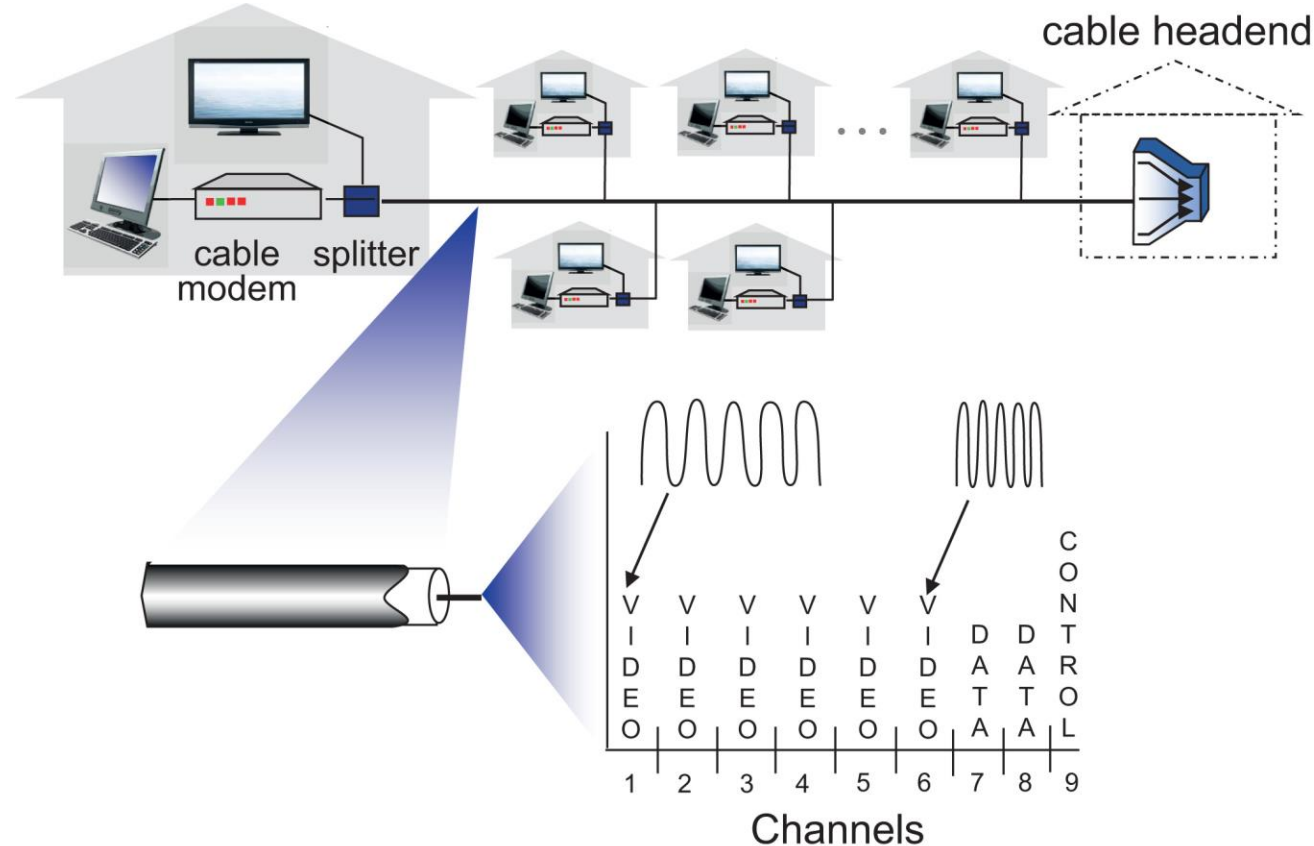
Access Networks and Physical Media

Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)

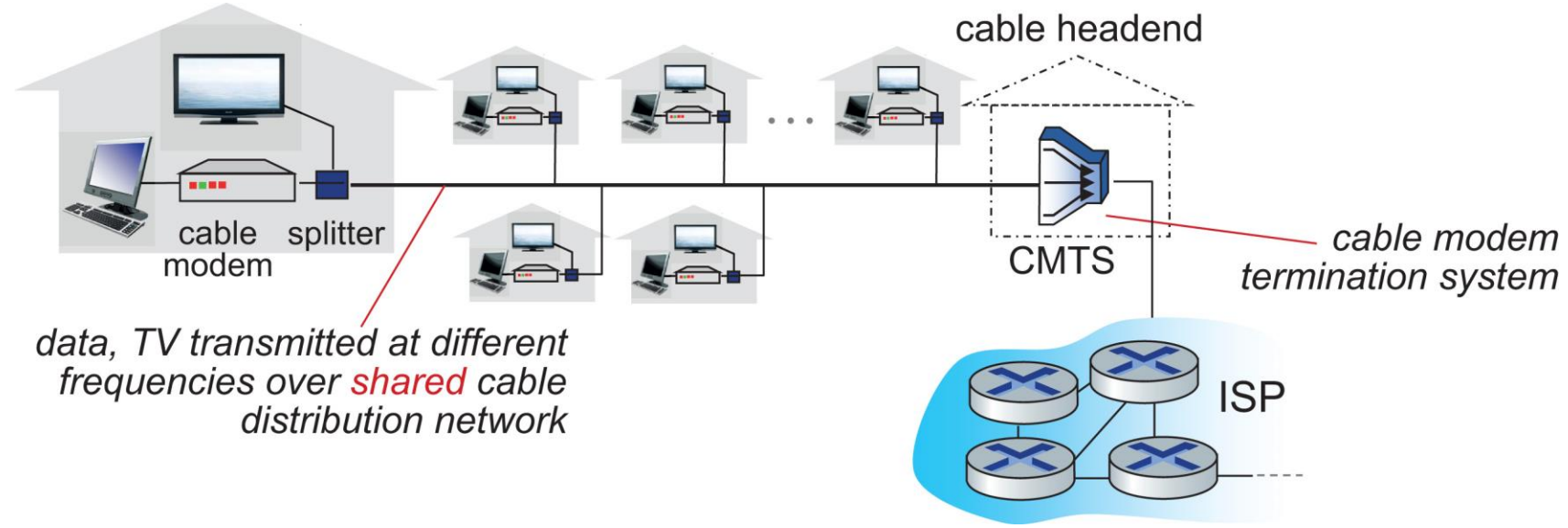


Access Networks: Cable-Based Access (1 of 2)



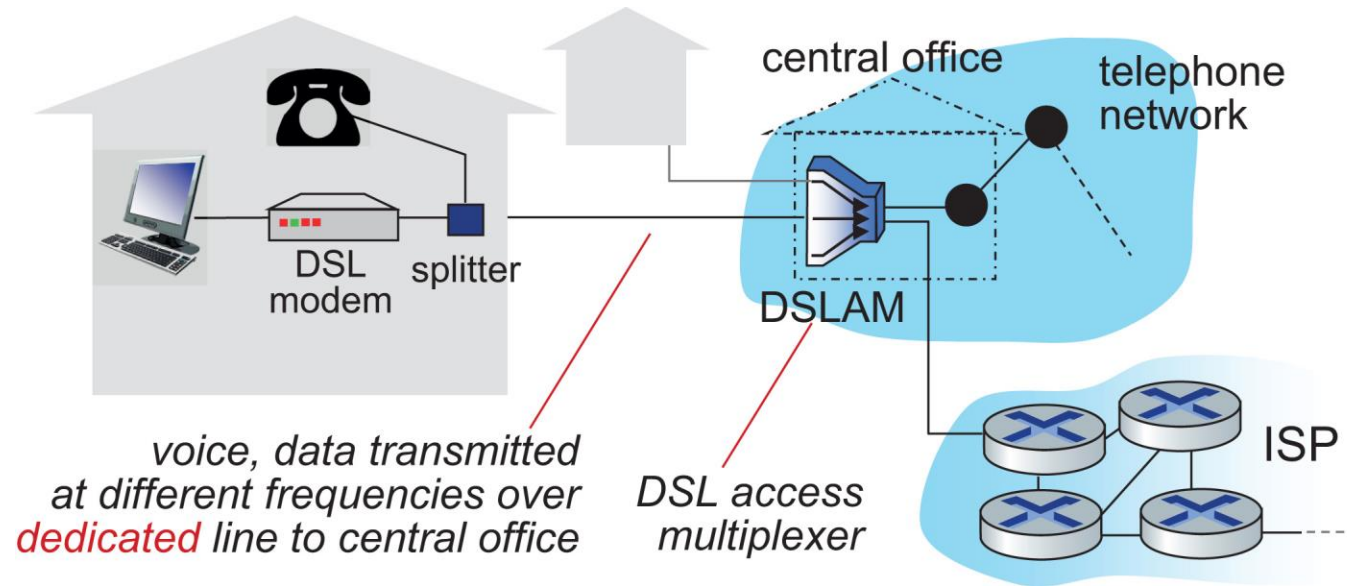
frequency division multiplexing (FDM): different channels transmitted in different frequency bands

Access Networks: Cable-Based Access (2 of 2)



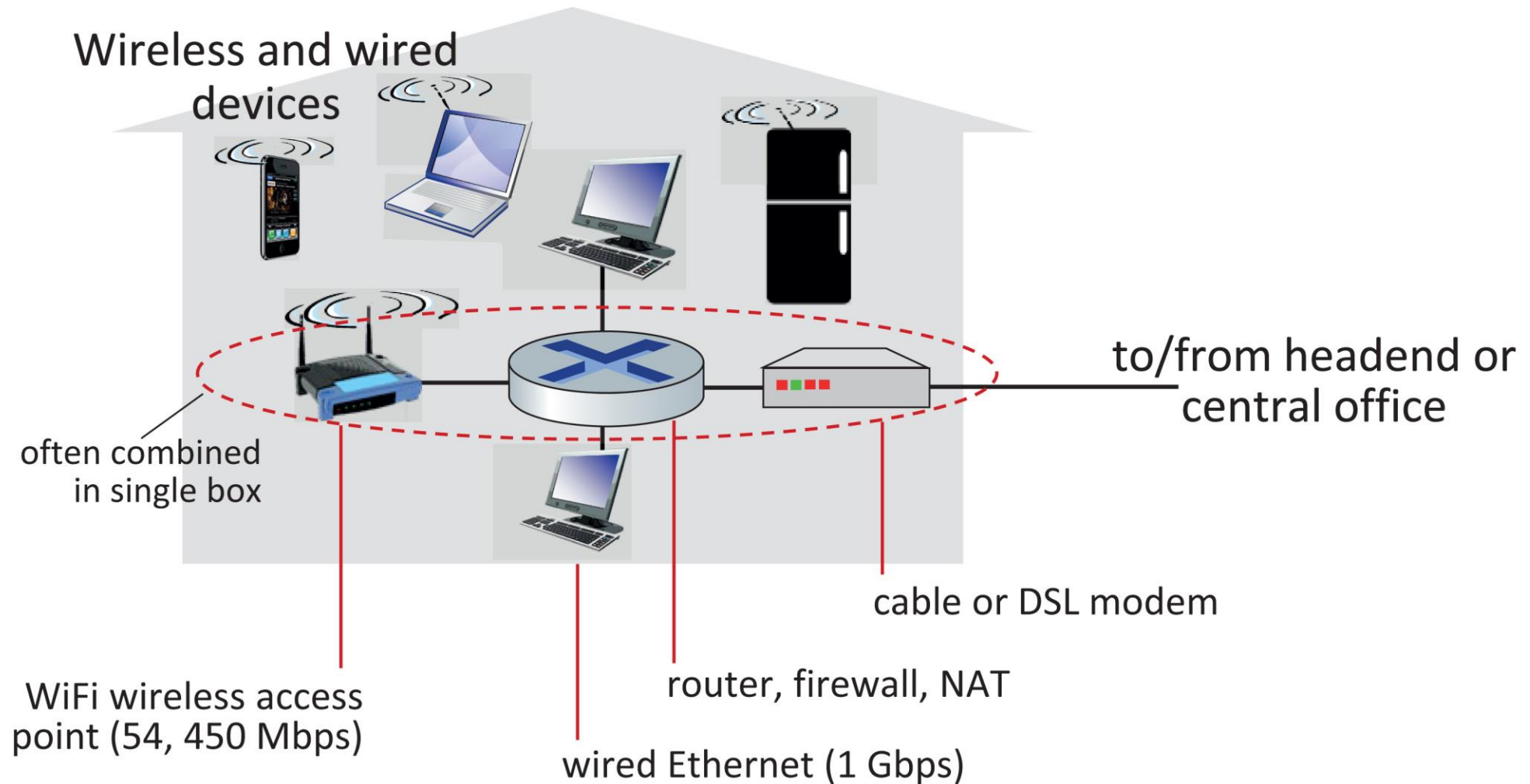
- HFC: hybrid fiber coax
 - asymmetric: up to 40 Mbps – 1.2 Gbps downstream transmission rate, 30-100 Mbps upstream transmission rate
- network of cable, fiber attaches homes to ISP router
 - homes **share access network** to cable headend

Access Networks: Digital Subscriber Line (DSL)



- use **existing** telephone line to central office DSLAM
 - data over DSL phone line goes to Internet
 - voice over DSL phone line goes to telephone net
- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate

Access Networks: Home Networks



Wireless Access Networks

Shared **wireless** access network connects end system to router

- via base station aka “access point”

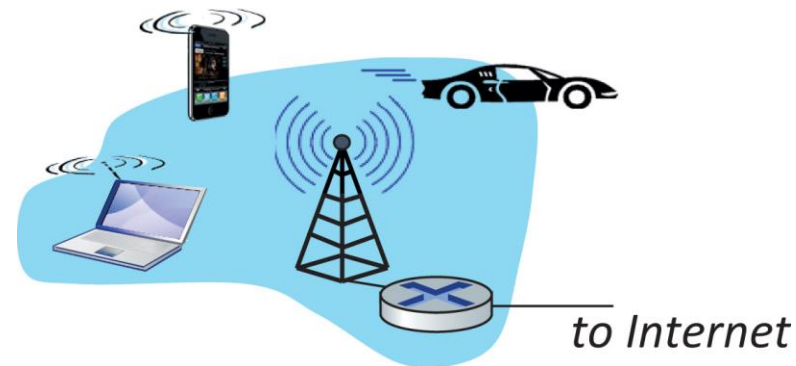
Wireless local area networks (WLANs)

- typically within or around building (~ 100 ft)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate

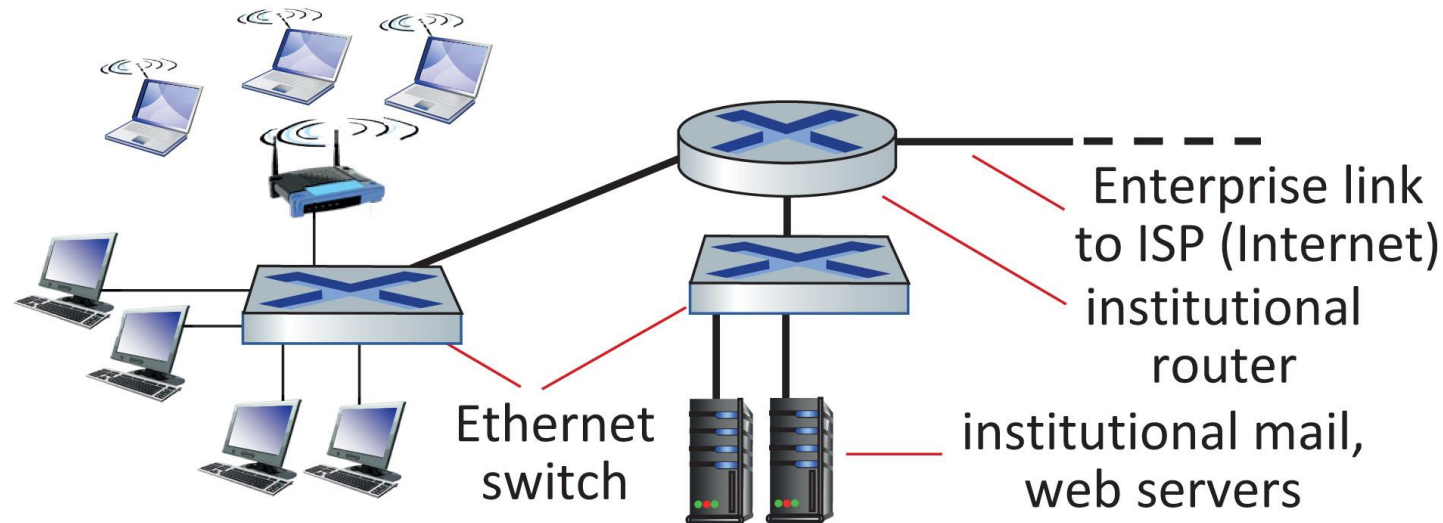


Wide-area cellular access networks

- provided by mobile, cellular network operator (10's km)
- 10's Mbps
- 4G/5G cellular networks



Access Networks: Enterprise Networks

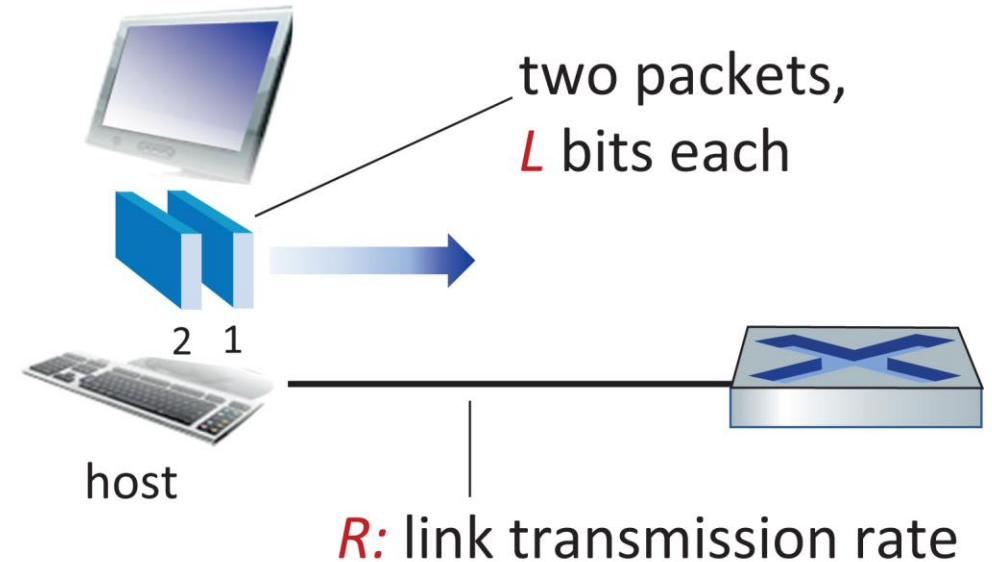


- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers (we'll cover differences shortly)
 - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
 - WiFi: wireless access points at 11, 54, 450 Mbps

Host: Sends Packets of Data

host sending function:

- takes application message
- breaks into smaller chunks, known as **packets**, of length L bits
- transmits packet into access network at **transmission rate R**
 - link transmission rate, aka link **capacity**, aka link **bandwidth**



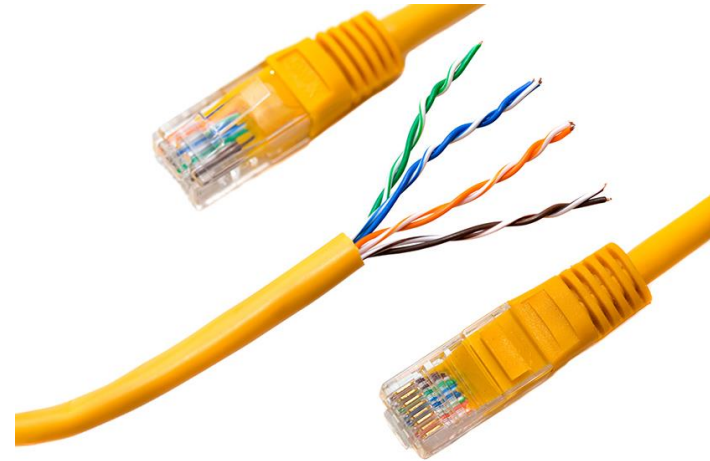
packet transmission delay = time needed to transmit L -bit packet into link = $\frac{L(\text{bits})}{R(\text{bits/sec})}$

Links: Physical Media (1 of 3)

- **bit**: propagates between transmitter/receiver pairs
- **physical link**: what lies between transmitter & receiver
- **guided media**:
 - signals propagate in solid media: copper, fiber, coax
- **unguided media**:
 - signals propagate freely, e.g., radio

Twisted pair (TP)

- two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps Ethernet



Links: Physical Media (2 of 3)

Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
 - multiple frequency channels on cable
 - 100's Mbps per channel



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
 - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



Links: Physical Media (3 of 3)

Wireless radio

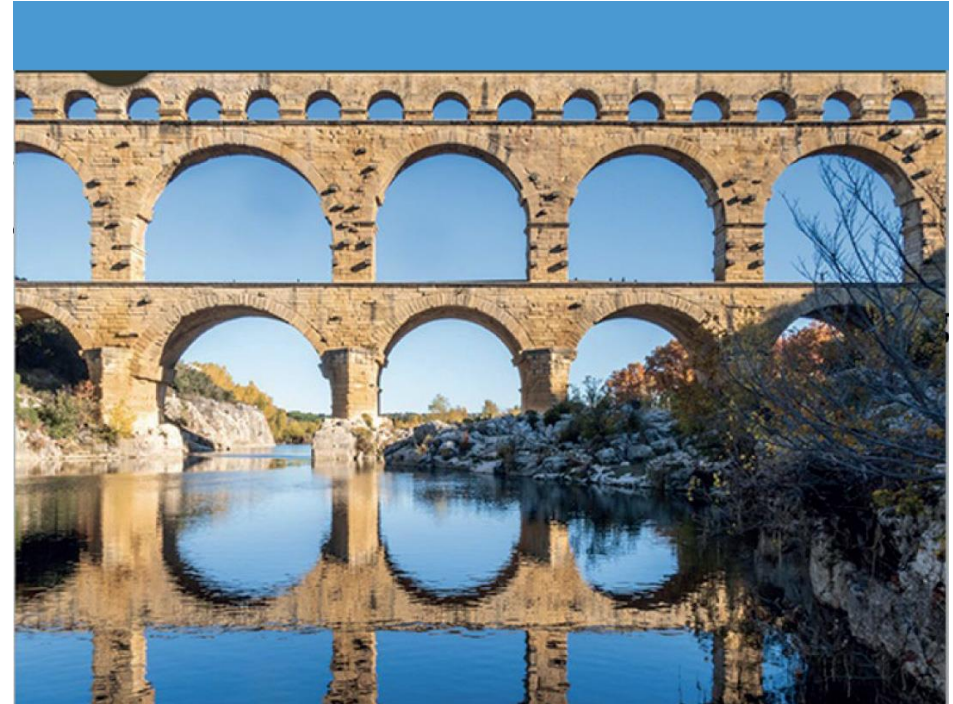
- signal carried in various “bands” in electromagnetic spectrum
- no physical “wire”
- broadcast, “half-duplex” (sender to receiver)
- propagation environment effects:
 - reflection
 - obstruction by objects
 - Interference/noise

Radio link types:

- **Wireless LAN (WiFi)**
 - 10-100’s Mbps; 10’s of meters
- **wide-area** (e.g., 4G/5G cellular)
 - 100’s Mbps (4G/5G) over ~10 Km
- **Bluetooth:** cable replacement
 - short distances, limited rates
- **terrestrial microwave**
 - point-to-point; 45 Mbps channels
- **satellite**
 - up to < 100 Mbps (Starlink) downlink
 - 270 msec end-end delay (geostationary)

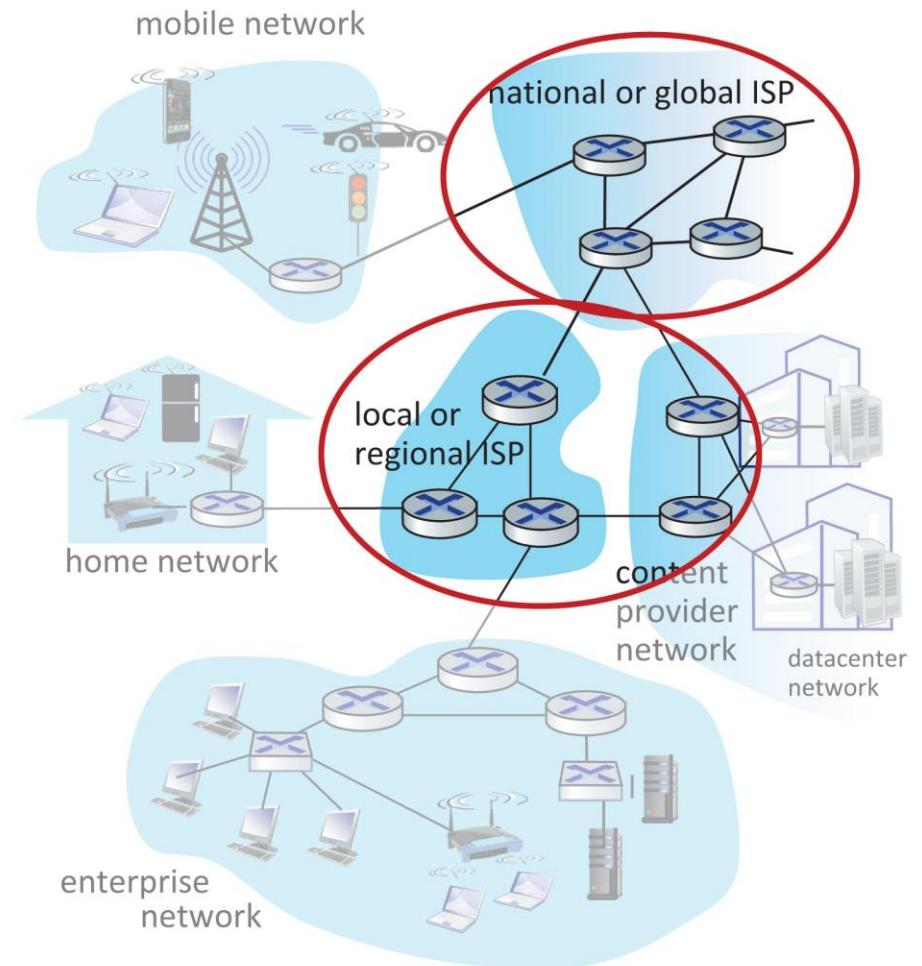
Chapter 1: Roadmap (2 of 6)

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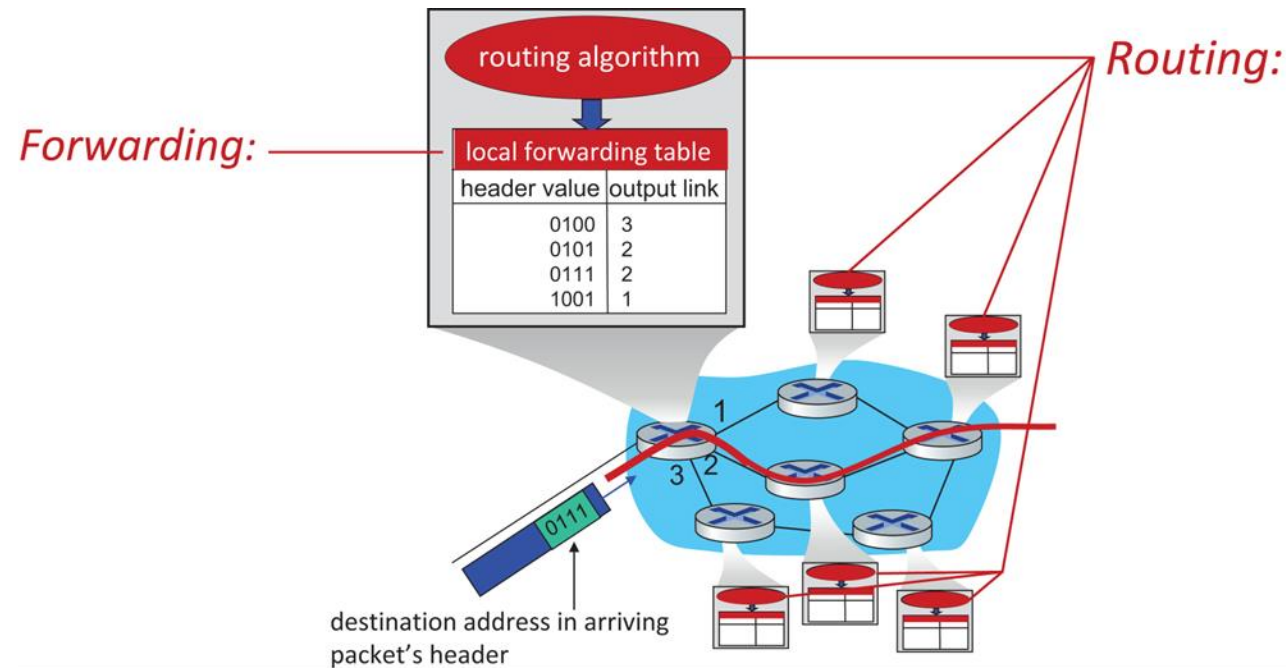
The Network Core

- mesh of interconnected routers
- **packet-switching**: hosts break application-layer messages into **packets**
 - network **forwards** packets from one router to the next, across links on path from **source to destination**



Two Key Network-Core Functions

- aka “switching”
- **local** action: move arriving packets from router’s input link to appropriate router output link

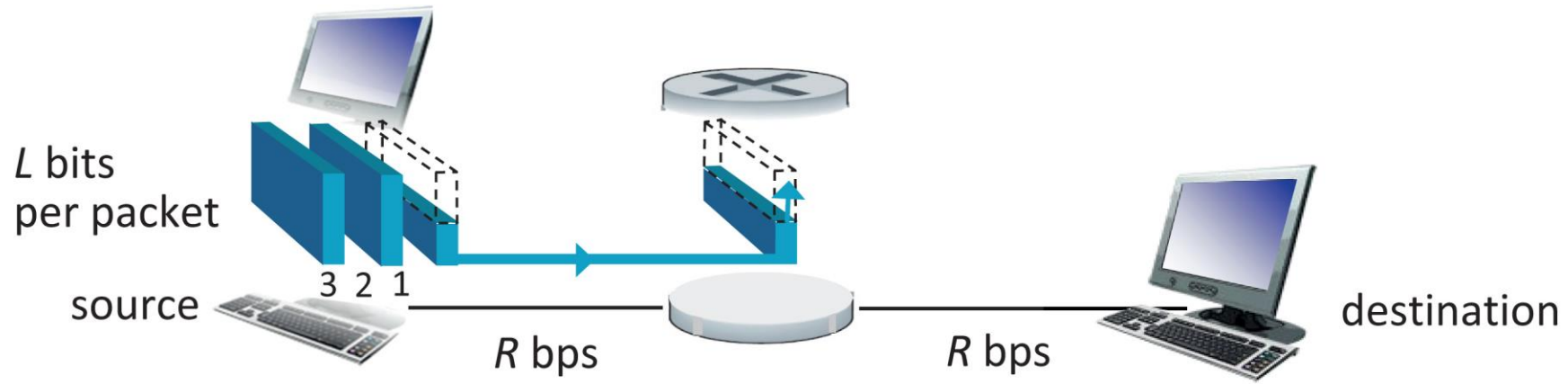


- **global** action: determine source-destination paths taken by packets
- routing algorithms





Packet-Switching: Store-and-Forward

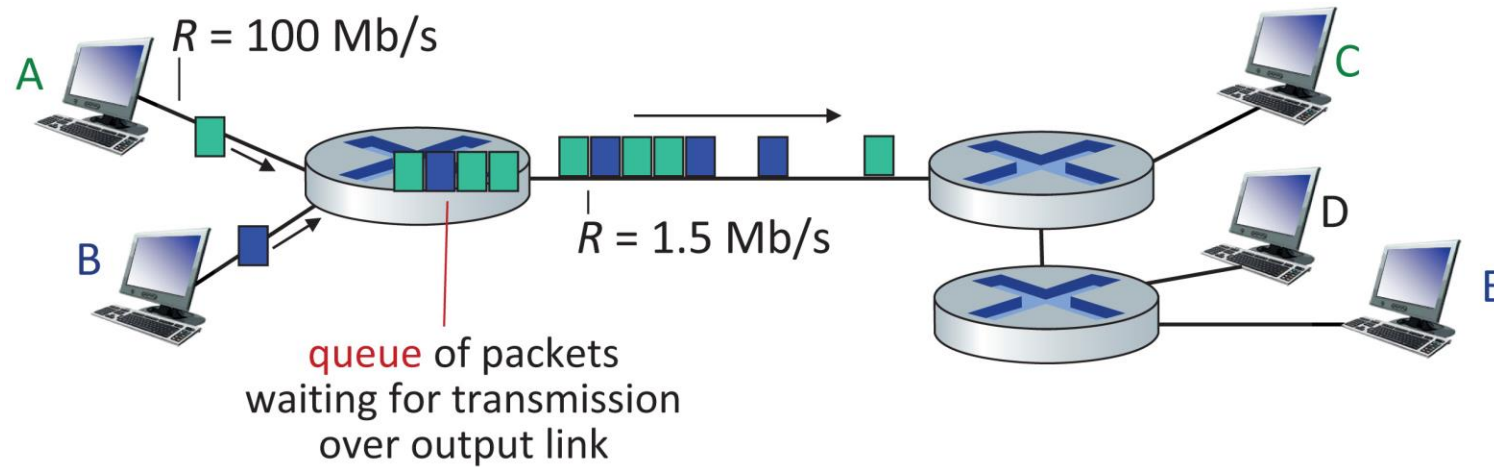


- **packet transmission delay:** takes L/R seconds to transmit (push out) L -bit packet into link at R bps
- **store and forward:** **entire** packet must arrive at router before it can be transmitted on next link

One-hop numerical example:

- $L = 10$ Kbits
- $R = 100$ M bps
- one-hop transmission delay = 0.1 msec

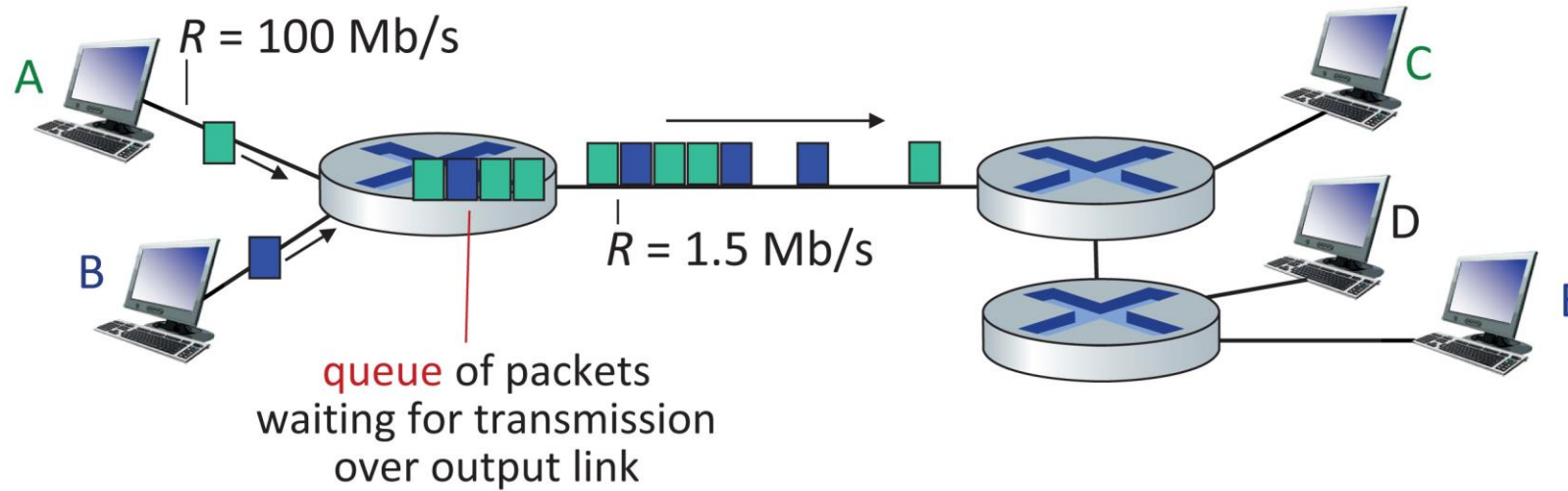
Packet-Switching: Queueing (1 of 2)



Queueing occurs when work arrives faster than it can be serviced:



Packet-Switching: Queueing (2 of 2)



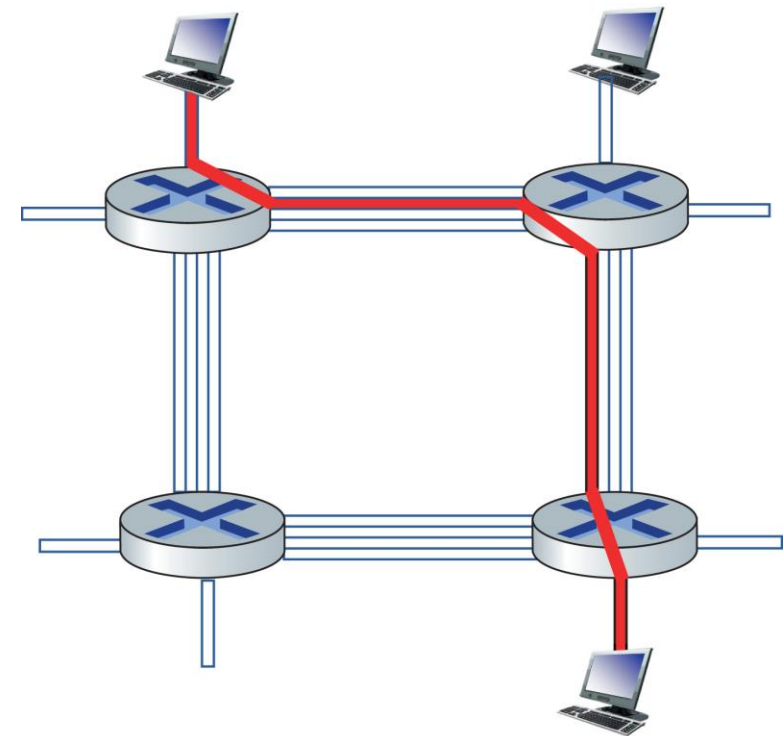
Packet queueing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time:

- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

Alternative to Packet Switching: Circuit Switching

end-end resources allocated to, reserved for “call” between source and destination

- in diagram, each link has four circuits.
 - call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
 - circuit-like (guaranteed) performance
- circuit segment idle if not used by call (no sharing)
- commonly used in traditional telephone networks



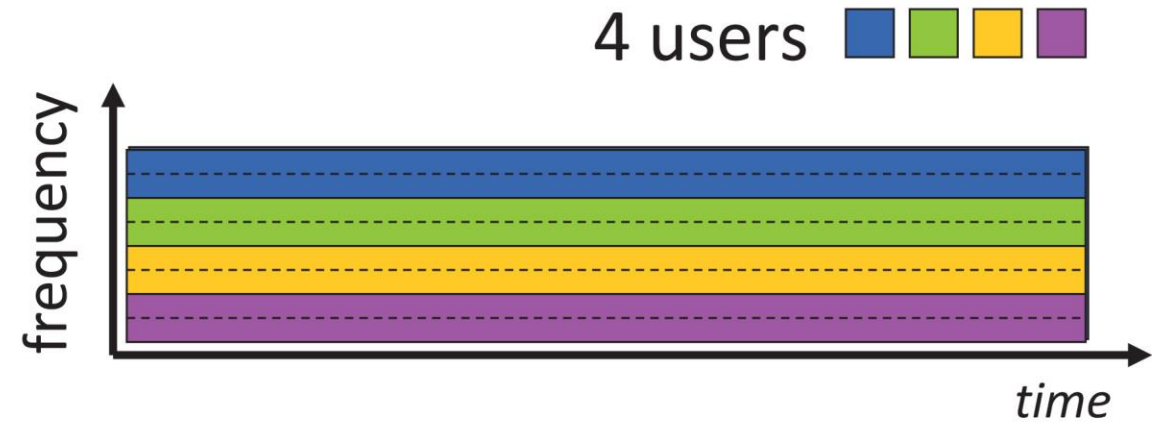
* Check out the online interactive exercises for more examples:

http://gaia.cs.umass.edu/kurose_ross/interactive

Circuit Switching: FDM and TDM

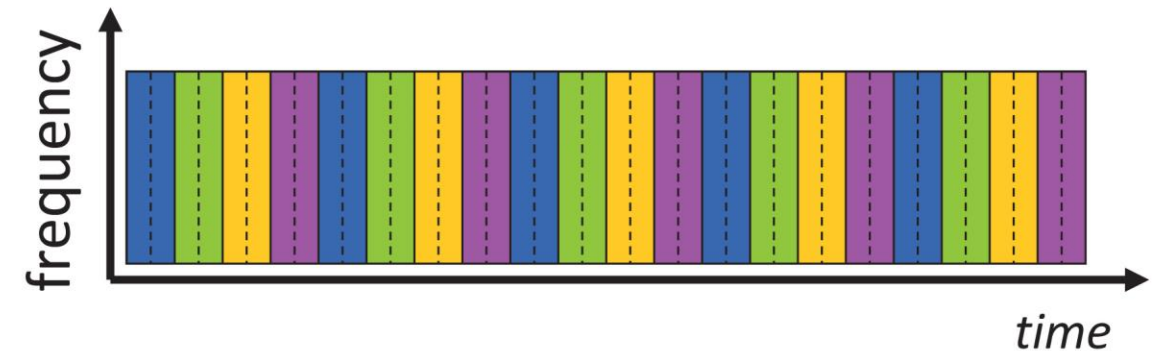
Frequency Division Multiplexing (FDM)

- optical, electromagnetic frequencies divided into (narrow) frequency bands
- each call allocated its own band, can transmit at max rate of that narrow band



Time Division Multiplexing (TDM)

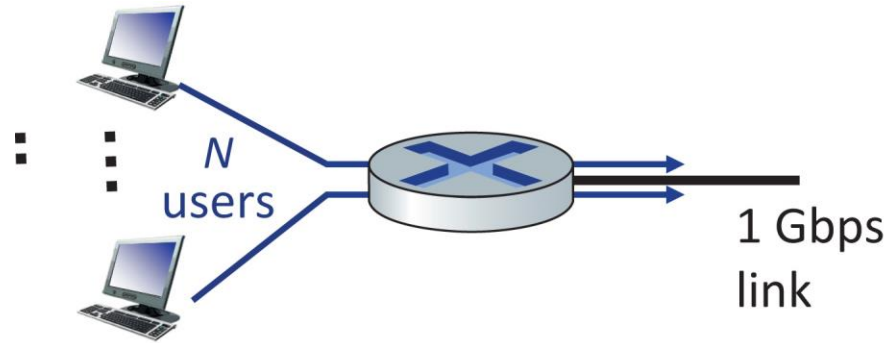
- time divided into slots
- each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band (only) during its time slot(s)



Packet Switching versus Circuit Switching (1 of 2)

example:

- 1 Gb/s link
- each user:
 - 100 Mb/s when “active”
 - active 10% of time



Q: how many users can use this network under circuit-switching and packet switching?

- **circuit-switching:** 10 users
- **packet switching:** with 35 users, probability > 10 active at same time is less than .0004 *

Q: how did we get value 0.0004?

A: HW problem (for those with course in probability only)

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Packet Switching versus Circuit Switching (2 of 2)

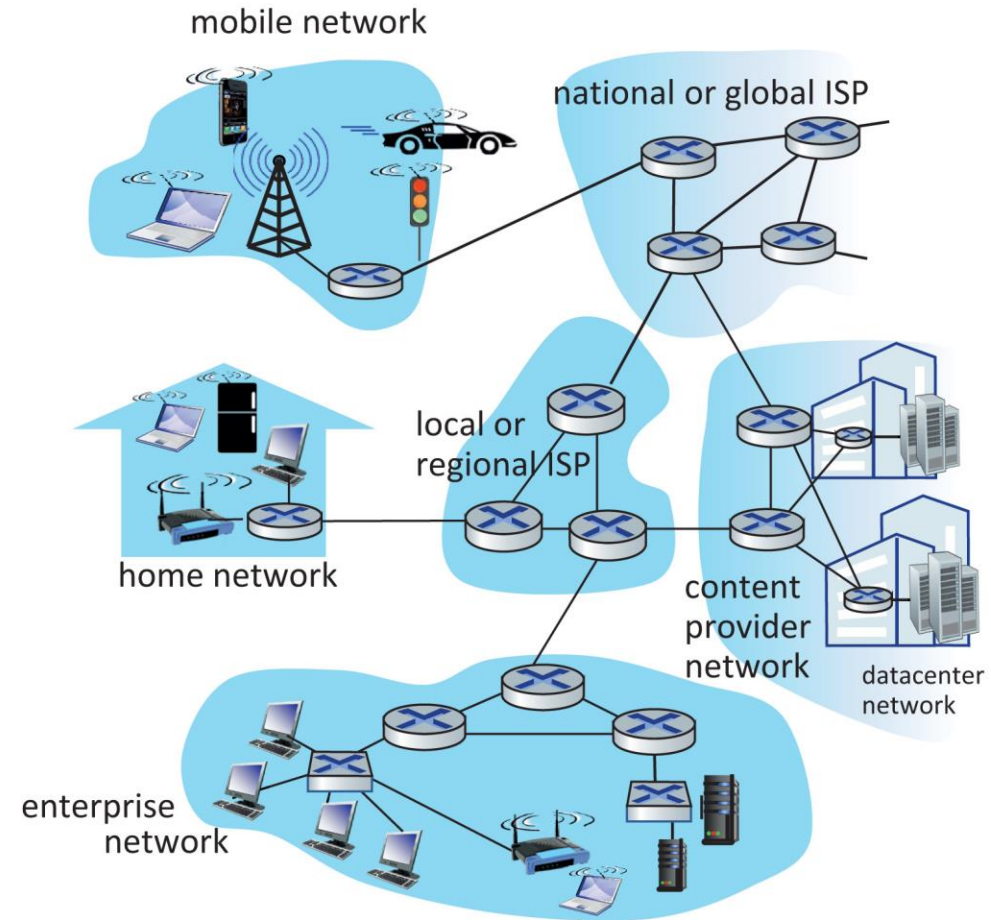
Is packet switching a “slam dunk winner”?

- great for “bursty” data – sometimes has data to send, but at other times not
 - resource sharing
 - simpler, no call setup
- **excessive congestion possible:** packet delay and loss due to buffer overflow
 - protocols needed for reliable data transfer, congestion control
- **Q: How to provide circuit-like behavior with packet-switching?**
 - “It’s complicated.” We’ll study various techniques that try to make packet switching as “circuit-like” as possible.

Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet switching)?

Internet Structure: a “Network of Networks” (1 of 9)

- hosts connect to Internet via **access** Internet Service Providers (ISPs)
- access ISPs in turn must be interconnected
 - so that **any** two hosts (**anywhere!**) can send packets to each other
- resulting network of networks is very complex
 - evolution driven by **economics**, **national policies**



Let's take a stepwise approach to describe current Internet structure

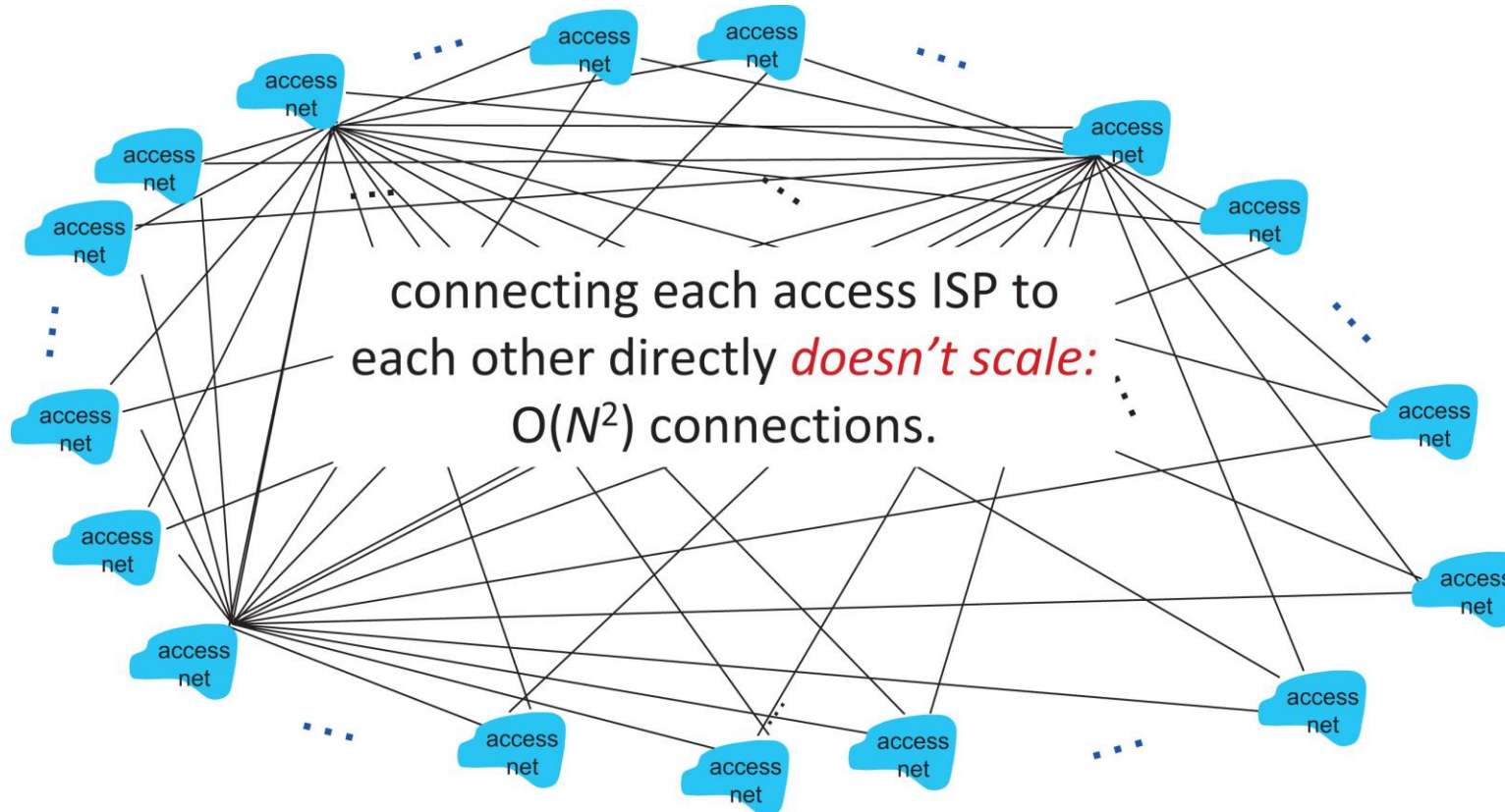
Internet Structure: a “Network of Networks” (2 of 9)

Question: given millions of access ISPs, how to connect them together?



Internet Structure: a “Network of Networks” (3 of 9)

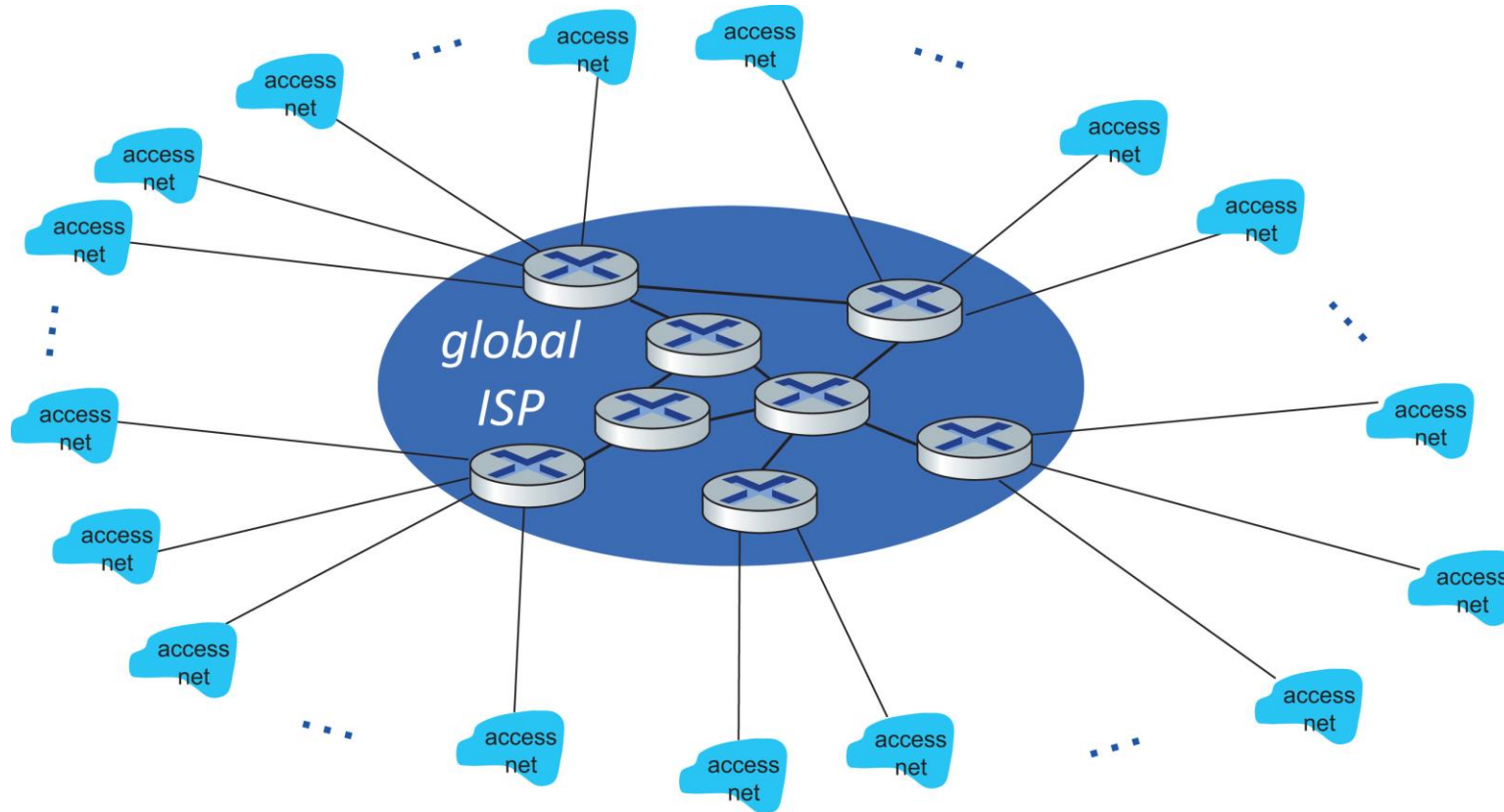
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Internet Structure: a “Network of Networks” (4 of 9)

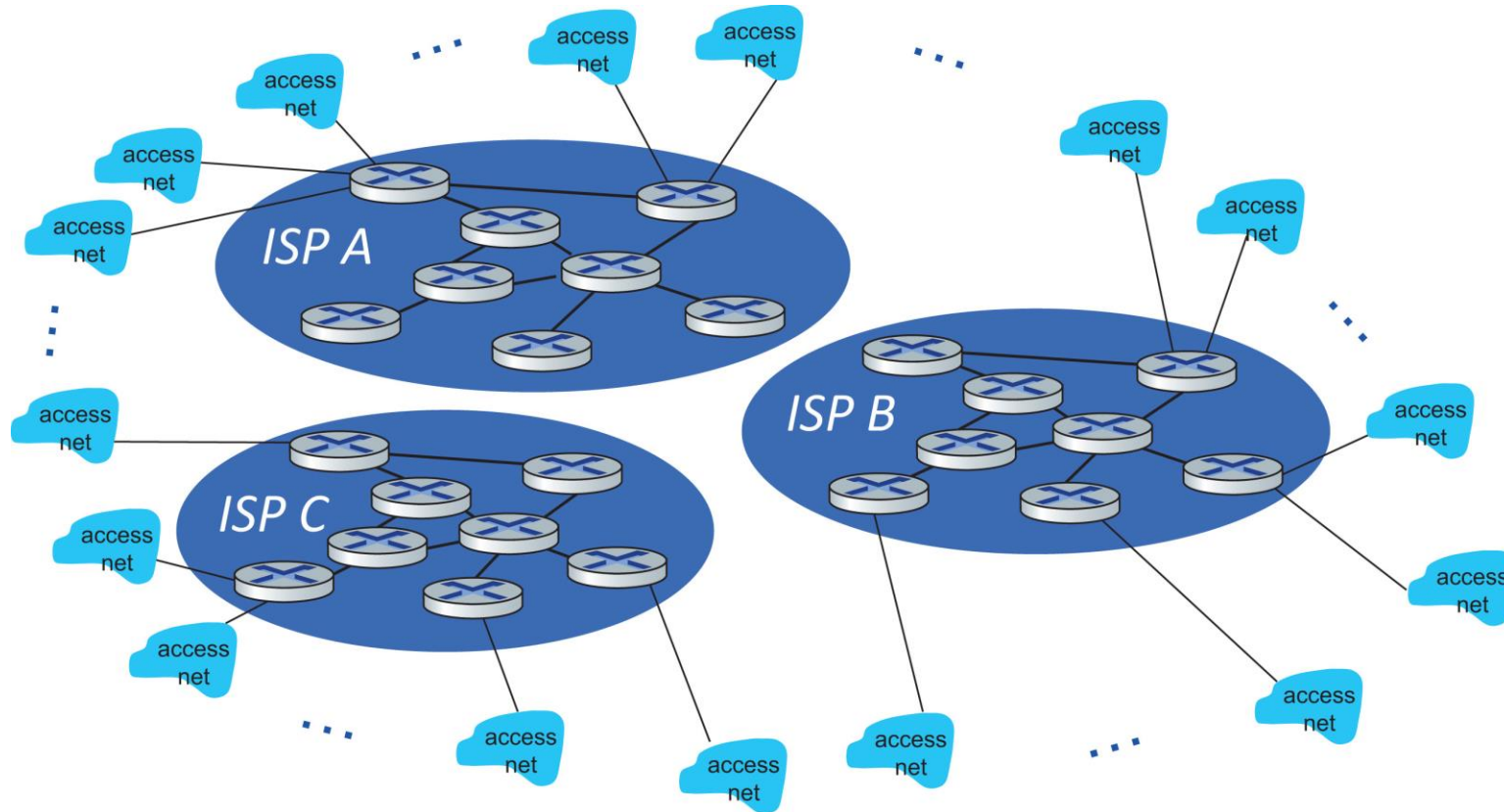
Option: connect each access ISP to one global transit ISP?

Customer and **provider** ISPs have economic agreement.



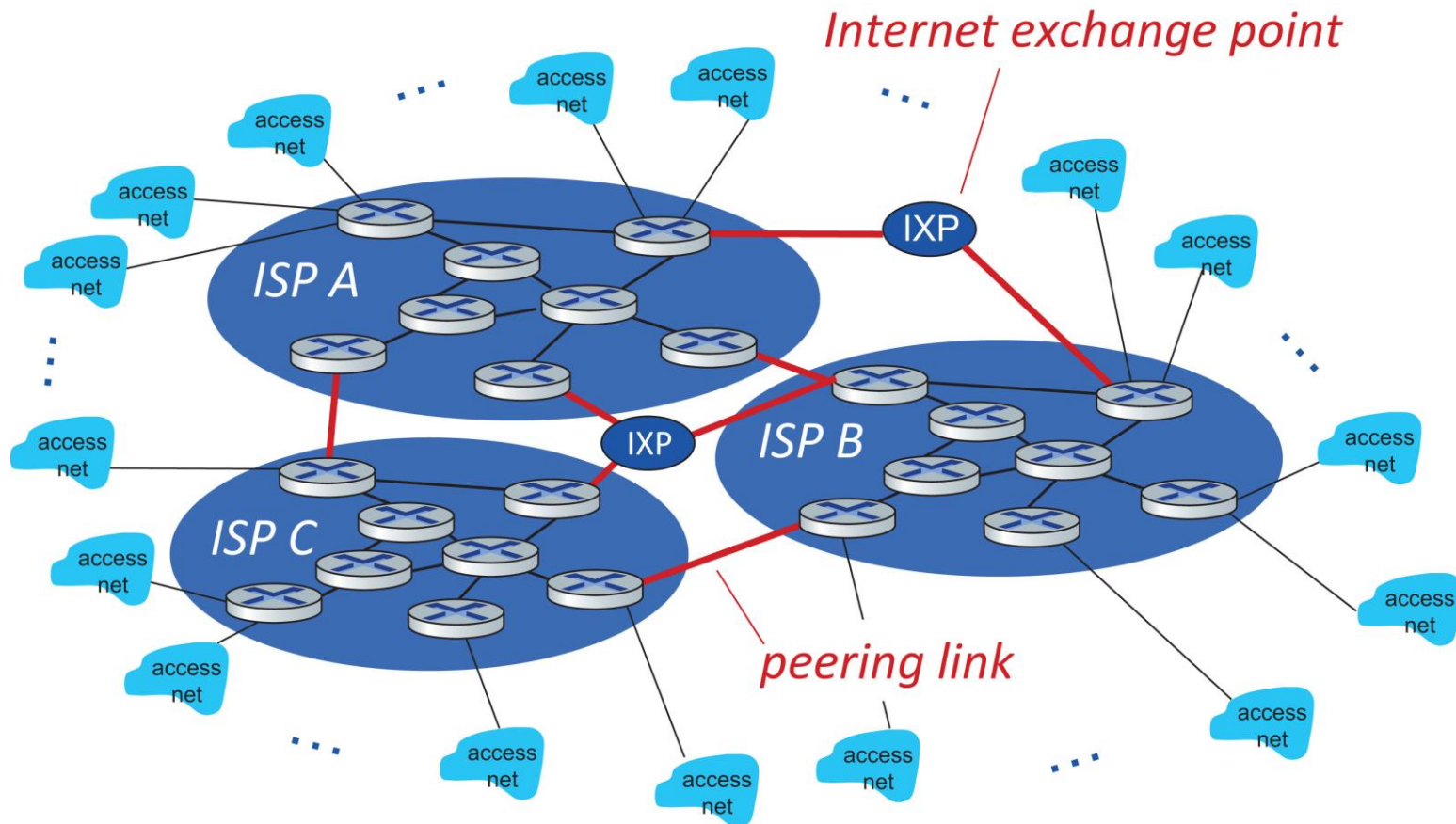
Internet Structure: a “Network of Networks” (5 of 9)

But if one global ISP is viable business, there will be competitors



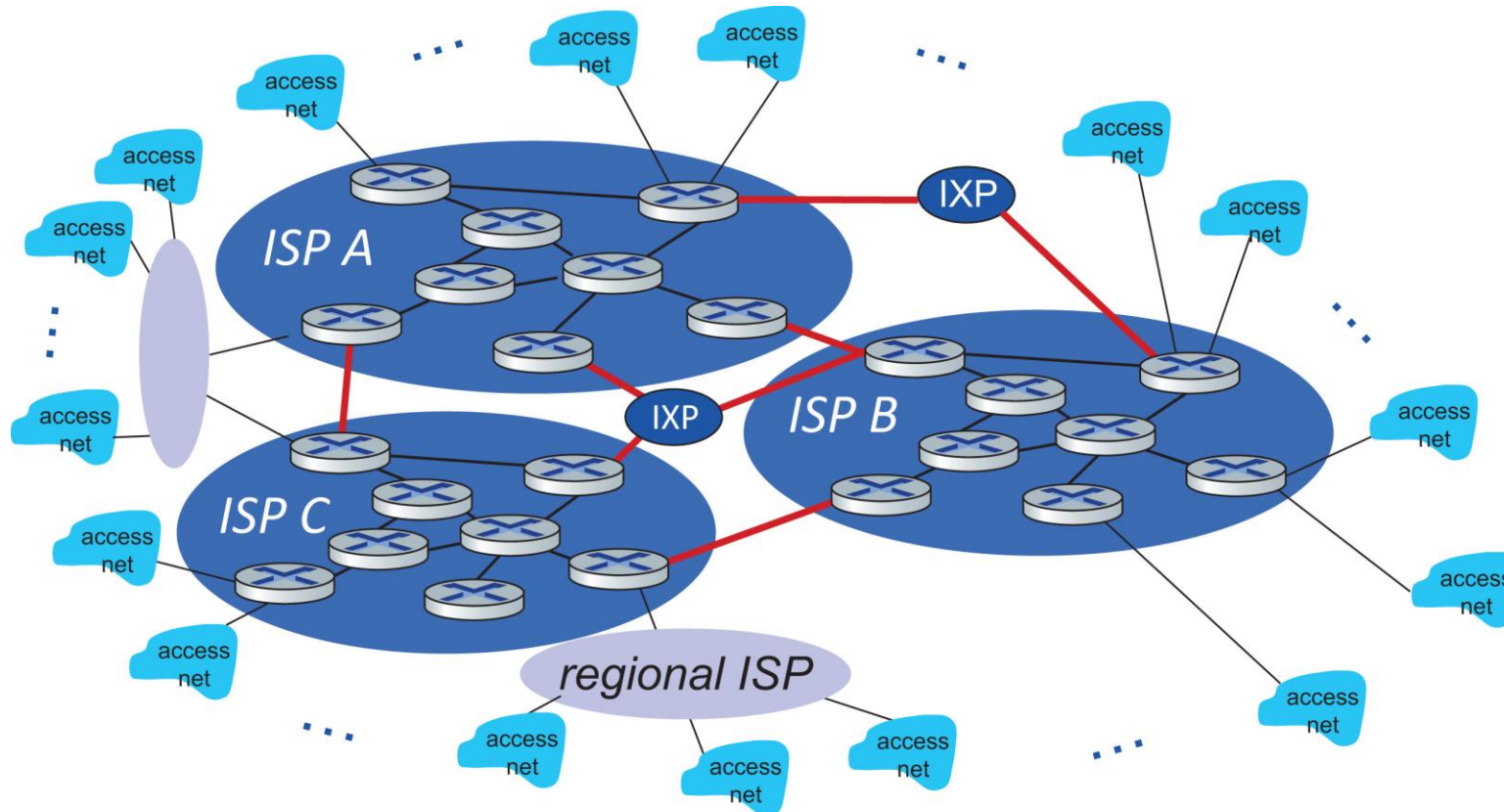
Internet Structure: a “Network of Networks” (6 of 9)

But if one global ISP is viable business, there will be competitors ... who will want to be connected



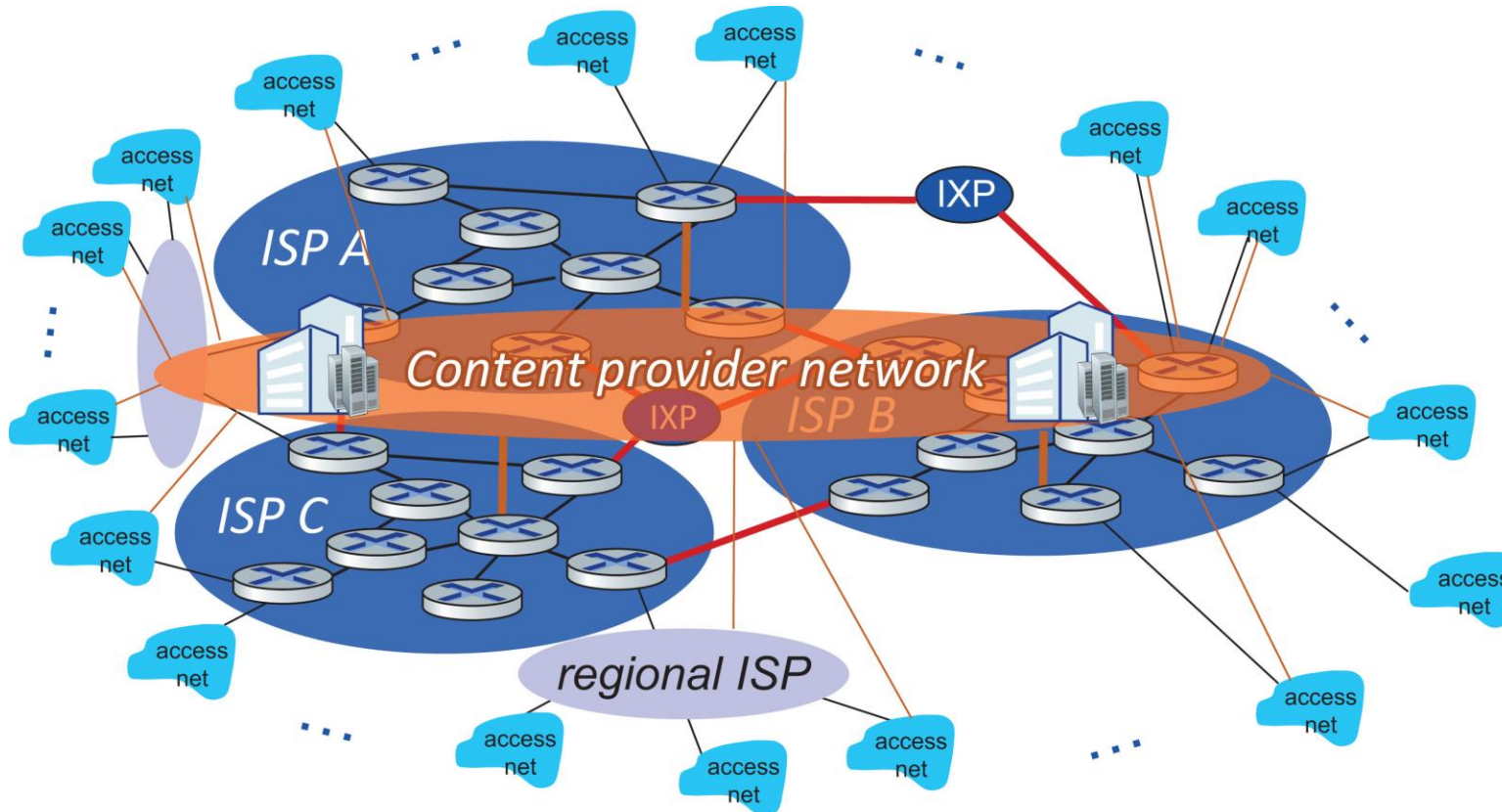
Internet Structure: a “Network of Networks” (7 of 9)

... and regional networks may arise to connect access nets to ISPs

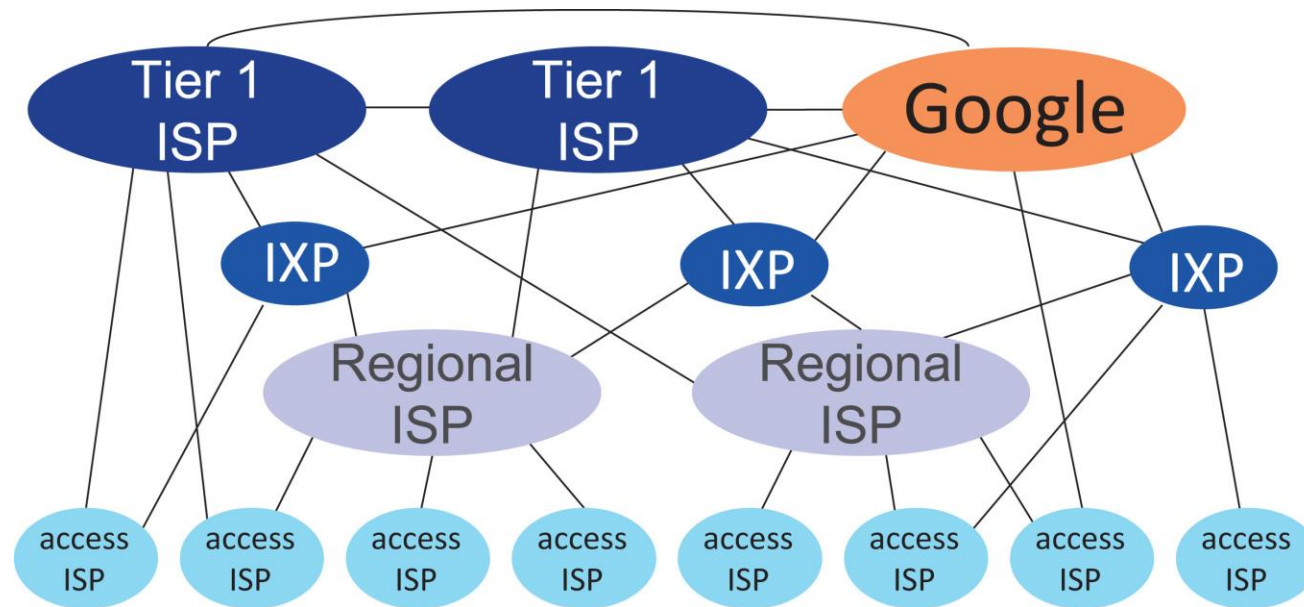


Internet Structure: a “Network of Networks” (8 of 9)

... and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet Structure: a “Network of Networks” (9 of 9)



At “center”: small # of well-connected large networks

- “tier-1” commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- content provider networks (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

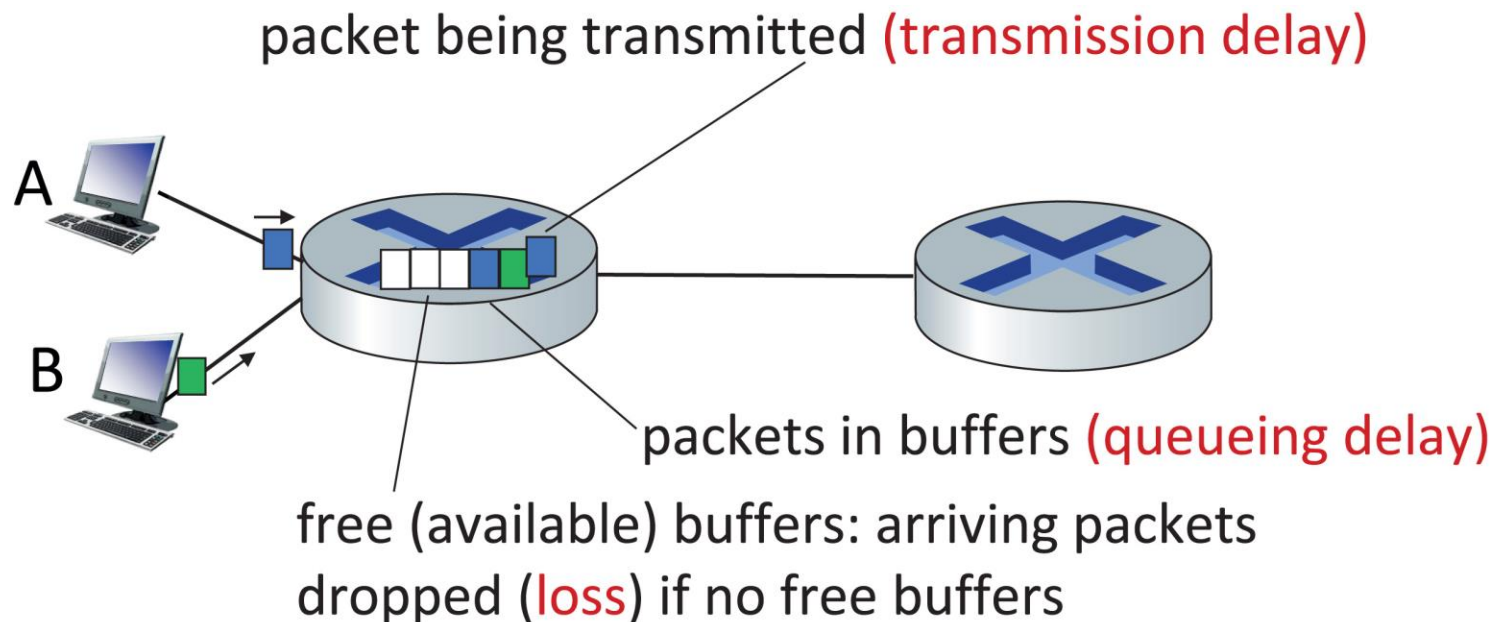
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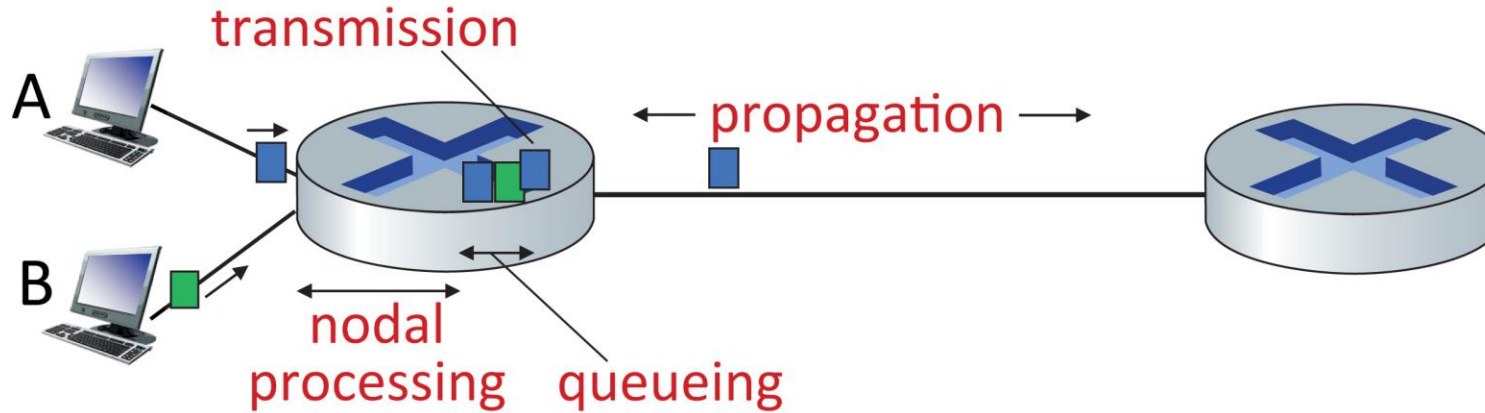


How do Packet Delay and Loss Occur?

- packets **queue** in router buffers, waiting for turn for transmission
 - queue length grows when arrival rate to link (temporarily) exceeds output link capacity
- packet **loss** occurs when memory to hold queued packets fills up



Packet Delay: Four Sources (1 of 2)



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

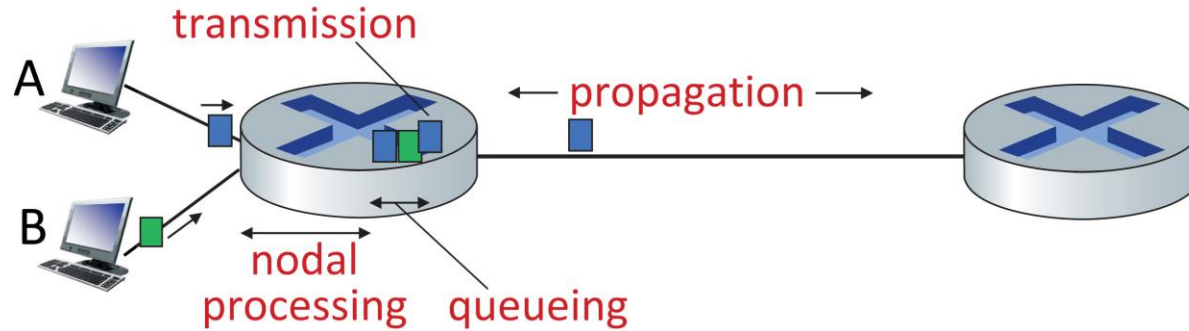
d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < microsecs

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Packet Delay: Four Sources (2 of 2)



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{trans} : transmission delay:

- L : packet length (bits)
- R : link transmission rate (bps)

d_{proc} : propagation delay:

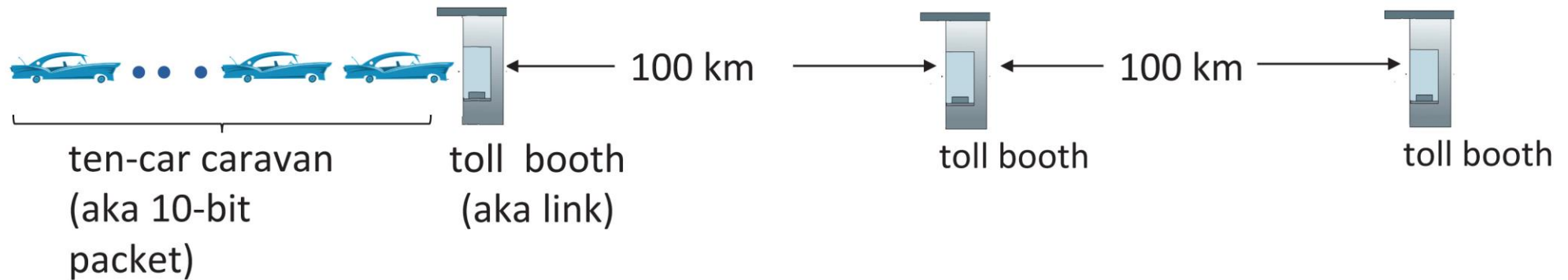
- d : length of physical link
- s : propagation speed ($\sim 2 \times 10^8$ m/sec)

• $d_{\text{trans}} = L/R$

• $d_{\text{proc}} = d/s$

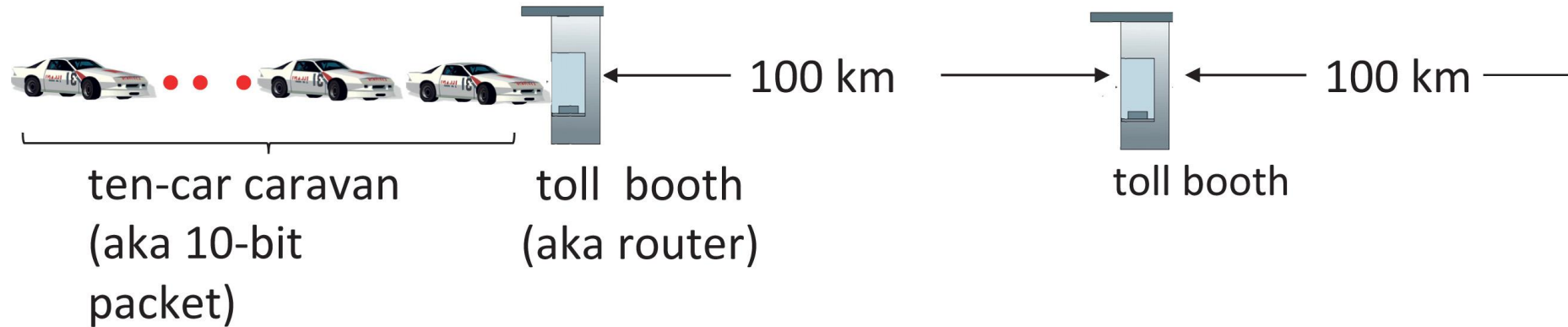
d_{trans} and d_{prop}
very different

Caravan Analogy (1 of 2)



- car ~ bit; caravan ~ packet;
toll service ~ link transmission
- toll booth takes 12 sec to service car (bit transmission time)
- “propagate” at 100 km/hr
- **Q: How long until caravan is lined up before 2nd toll booth?**
- time to “push” entire caravan through toll booth onto highway
 $= 12 * 10 = 120 \text{ sec}$
- time for last car to propagate from 1st to 2nd toll booth:
 $100 \text{ km} / (100 \text{ km} / \text{hr}) = 1 \text{ hr}$
- **A: 62 minutes**

Caravan Analogy (2 of 2)



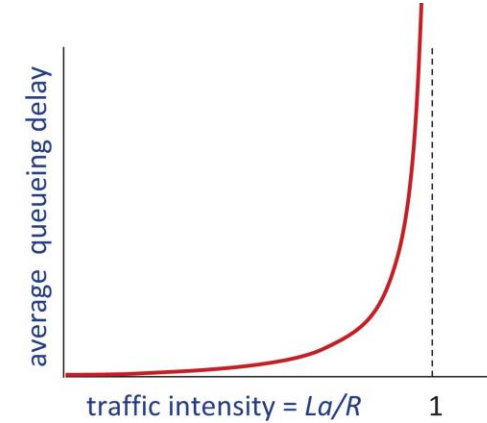
- suppose cars now “propagate” at 1000 km/hr
 - and suppose toll booth now takes one min to service a car
 - **Q: Will cars arrive to 2nd booth before all cars serviced at first booth?**
- A: Yes!** after 7 min, first car arrives at second booth; three cars still at first booth

Packet Queueing Delay (Revisited)

- a : average packet arrival rate
- L : packet length (bits)
- R : link bandwidth (bit transmission rate)

$$\frac{L \cdot a}{R} : \frac{\text{arrival rate of bits}}{\text{service rate of bits}} \quad \text{“traffic intensity”}$$

- $La/R \sim 0$: avg. queueing delay small
- $La/R \rightarrow 1$: avg. queueing delay large
- $La/R > 1$: more “work” arriving is more than can be serviced – average delay infinite!



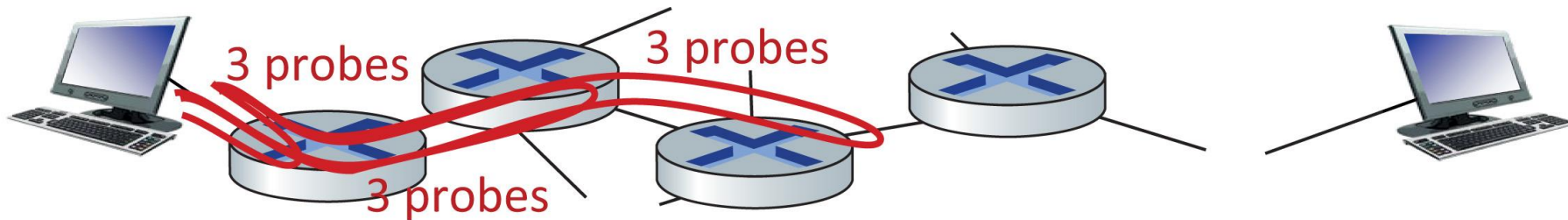
$La/R \sim 0$



$La/R \rightarrow 1$

“Real” Internet Delays and Routes

- what do “real” Internet delay & loss look like?
- **traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination (with time-to-live field value of i)
 - router i will return packets to sender
 - sender measures time interval between transmission and reply



Real Internet Delays and Routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

3 delay measurements
to border1-rt-fa5-1-0.gw.umass.edu

trans-oceanic link

looks like delays
decrease! Why?

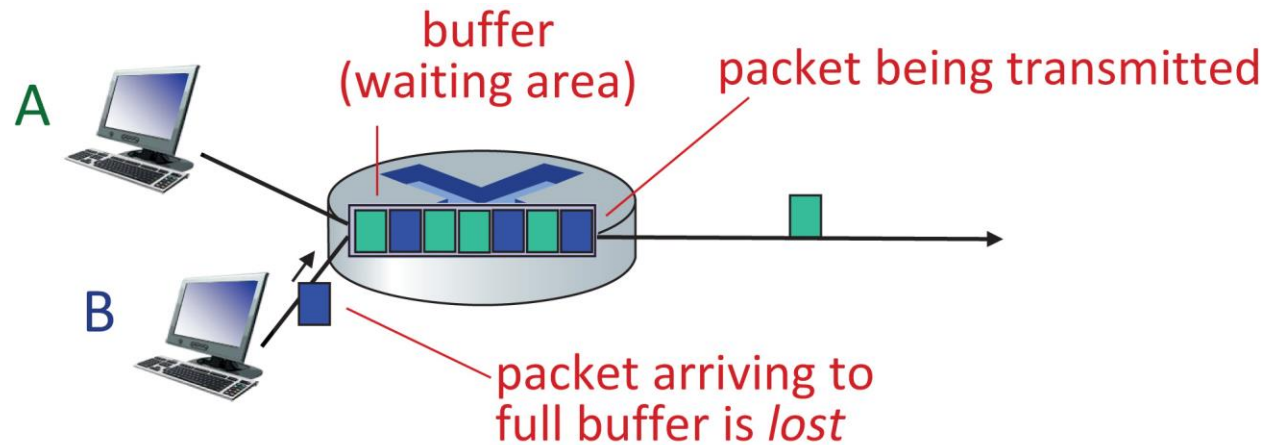
* means no response (probe lost, router not replying)

```
1 cs-gw (128.119.240.254) 1 ms 1 ms 2 ms
2 border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145) 1 ms 1 ms 2 ms
3 cht-vbns.gw.umass.edu (128.119.3.130) 6 ms 5 ms 5 ms
4 jn1-at1-0-0-19.wor.vbns.net (204.147.132.129) 16 ms 11 ms 13 ms
5 jn1-so7-0-0-0.wae.vbns.net (204.147.136.136) 21 ms 18 ms 18 ms
6 abilene-vbns.abilene.ucaid.edu (198.32.11.9) 22 ms 18 ms 22 ms
7 nycm-wash.abilene.ucaid.edu (198.32.8.46) 22 ms 22 ms 22 ms
8 62.40.103.253 (62.40.103.253) 104 ms 109 ms 106 ms
9 de2-1.de1.de.geant.net (62.40.96.129) 109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50) 113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54) 112 ms 114 ms 112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13) 111 ms 114 ms 116 ms
13 nice.cssi.renater.fr (195.220.98.102) 123 ms 125 ms 124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110) 126 ms 126 ms 124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54) 135 ms 128 ms 133 ms
16 194.214.211.25 (194.214.211.25) 126 ms 128 ms 126 ms
17 * * *
18 * * *
19 fantasia.eurecom.fr (193.55.113.142) 132 ms 128 ms 136 ms
```

* Do some traceroutes from exotic countries at www.traceroute.org

Packet Loss

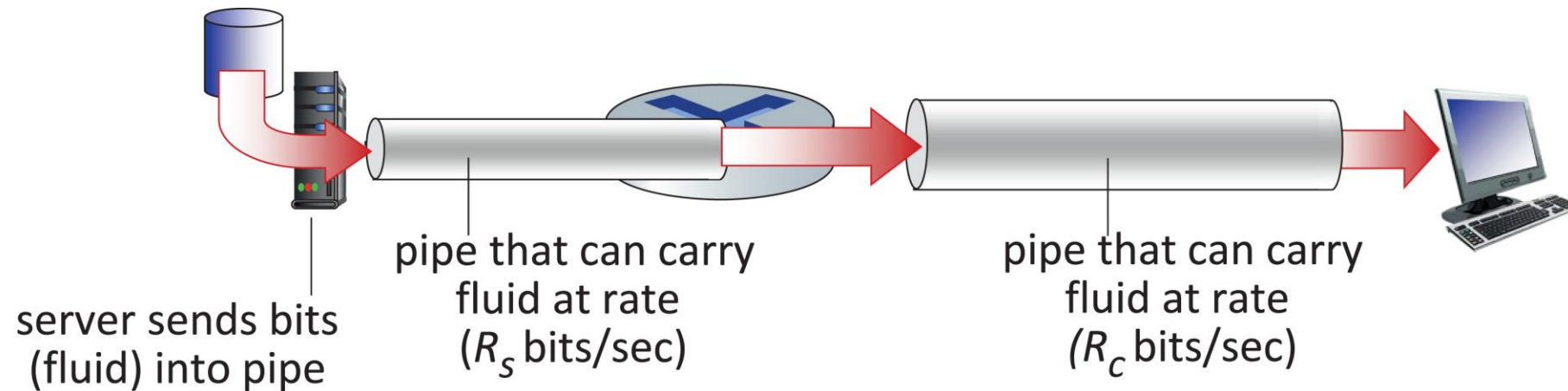
- queue (aka buffer) preceding link in buffer has finite capacity
- packet arriving to full queue dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all



* Check out the Java applet for an interactive animation (on publisher's website) of queuing and loss

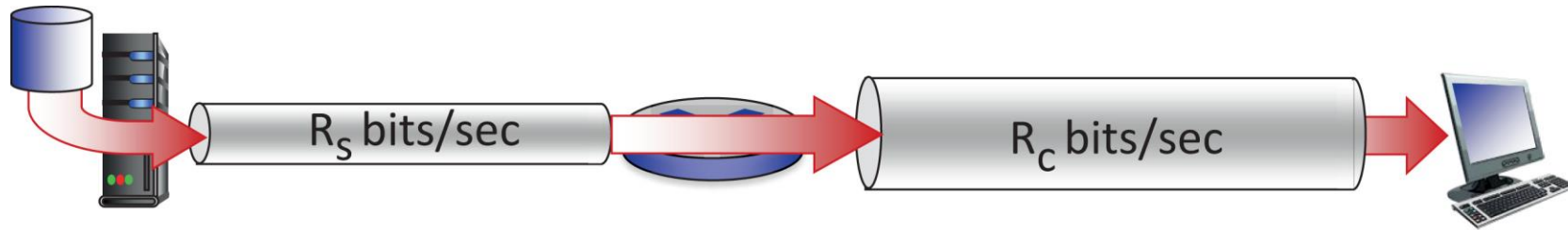
Throughput (1 of 2)

- **throughput:** rate (bits/time unit) at which bits are being sent from sender to receiver
 - **instantaneous:** rate at given point in time
 - **average:** rate over longer period of time

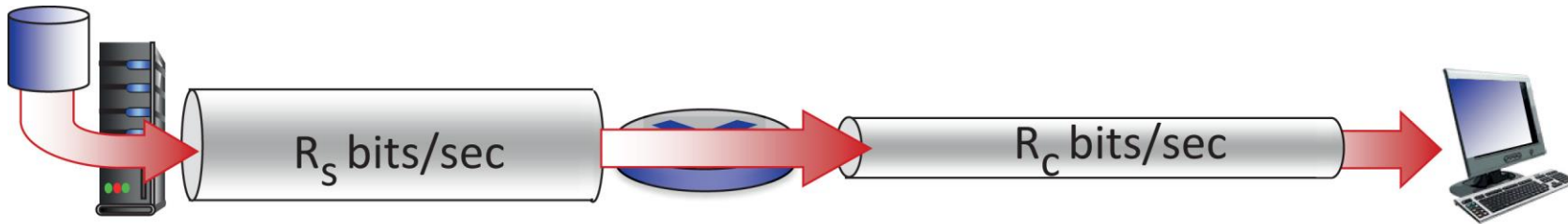


Throughput (2 of 2)

$R_s < R_c$ What is average end-end throughput?



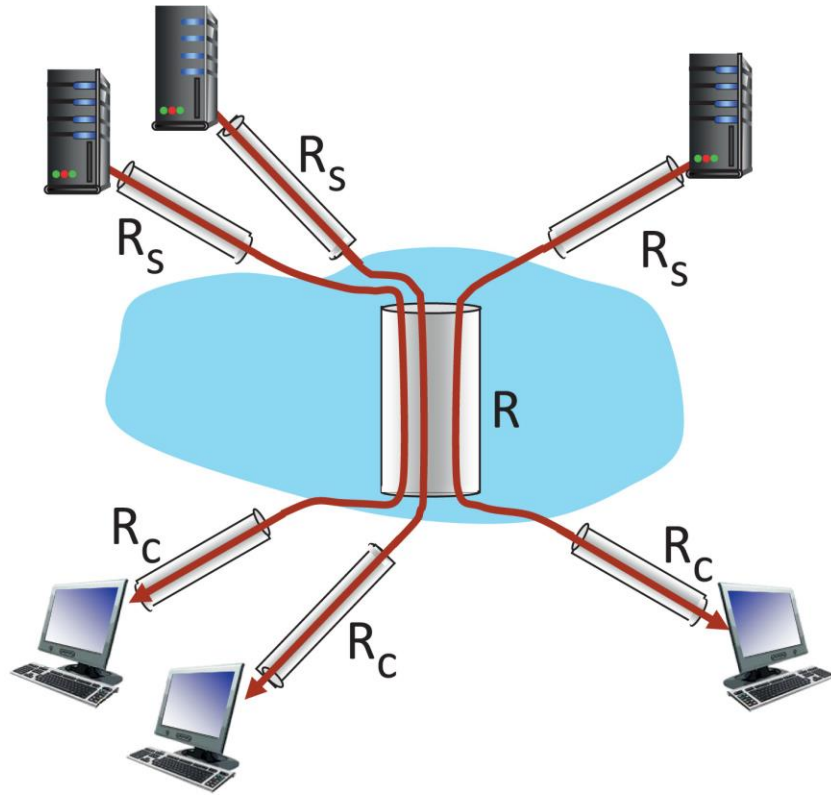
$R_s > R_c$ What is average end-end throughput?



bottleneck link

link on end-end path that constrains end-end throughput

Throughput: Network Scenario



10 connections (fairly) share
backbone bottleneck link R bits/sec

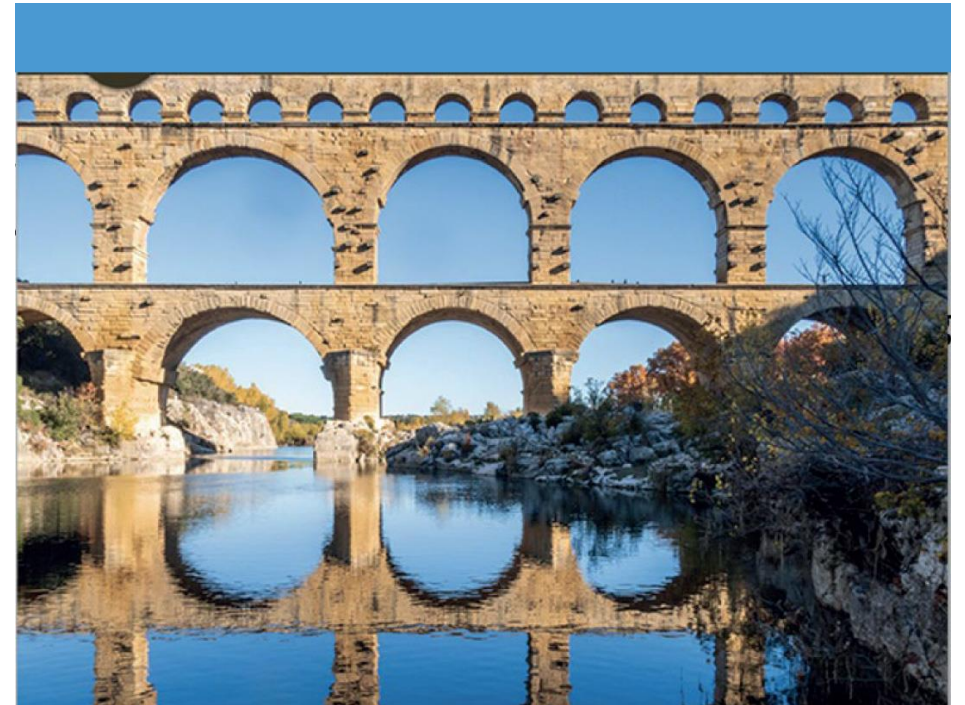
- per-connection end-end throughput:
 $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck

* Check out the online interactive exercises for more examples:

http://gaia.cs.umass.edu/kurose_ross/

Chapter 1: Roadmap (5 of 6)

- What is the Internet?
- What is a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- Protocol layers, service models
- History



Protocol “Layers” and Reference Models

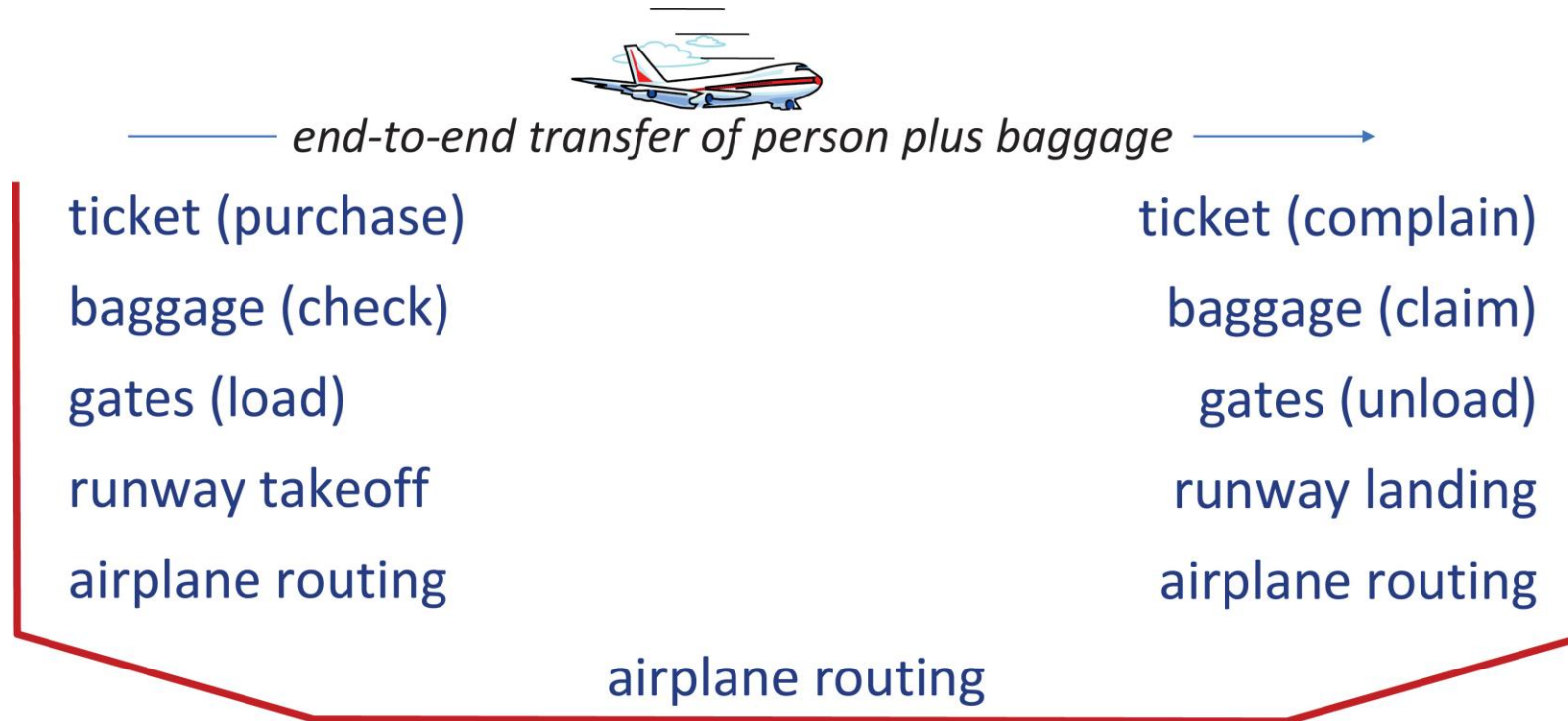
Networks are complex, with many “pieces”:

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question: is there any hope of organizing structure of network?

- and/or our **discussion** of networks?

Example: Organization of Air Travel (1 of 2)



How would you **define/discuss** the *system* of airline travel?

- a series of steps, involving many services

Example: Organization of Air Travel (2 of 2)



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

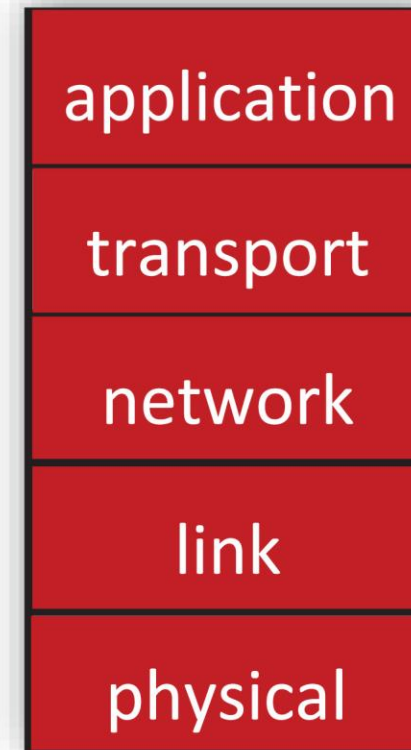
Why Layering?

Approach to designing/discussing complex systems:

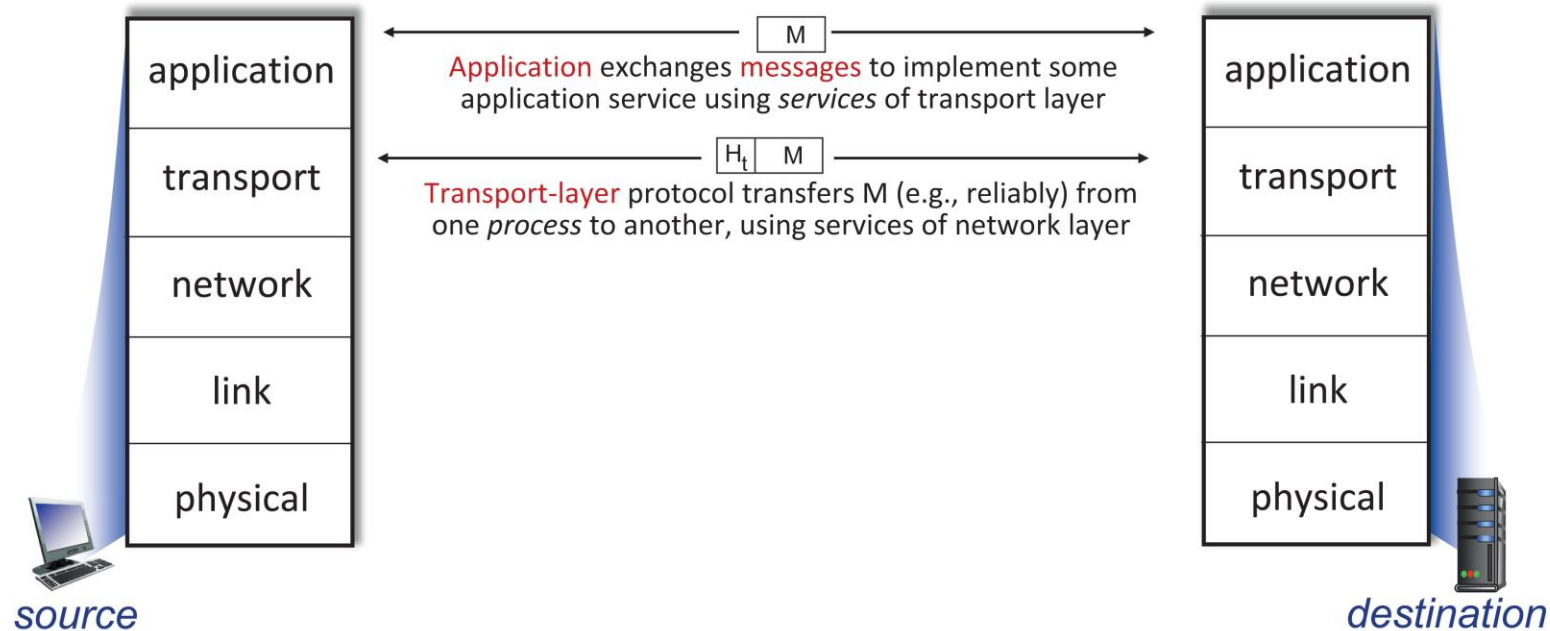
- explicit structure allows identification, relationship of system's pieces
 - layered **reference model** for discussion
- modularization eases maintenance, updating of system
 - change in layer's service **implementation**: transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system

Layered Internet Protocol Stack

- **application:** supporting network applications
 - HTTP, IMAP, SMTP, DNS
- **transport:** process-process data transfer
 - TCP, UDP
- **network:** routing of datagrams from source to destination
 - IP, routing protocols
- **link:** data transfer between neighboring network elements
 - Ethernet, 802.11 (WiFi), PPP
- **physical:** bits “on the wire”

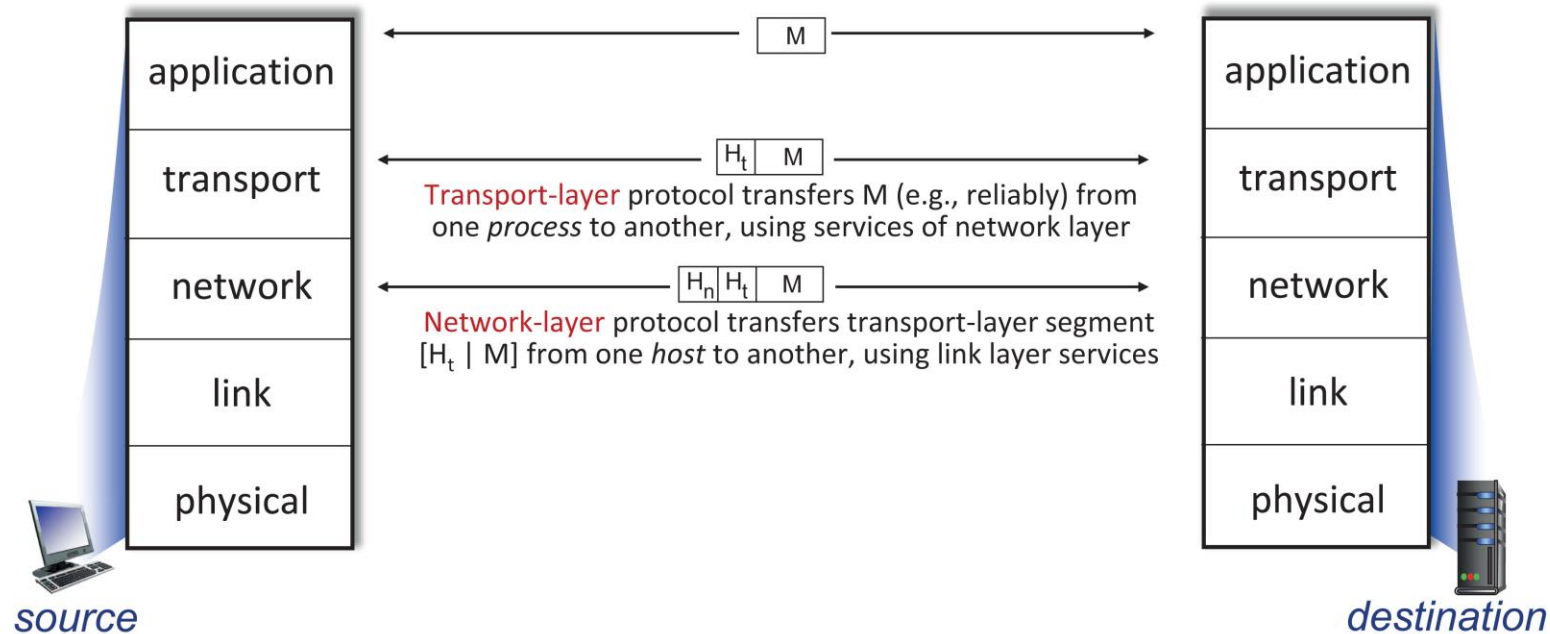


Services, Layering and Encapsulation (1 of 5)



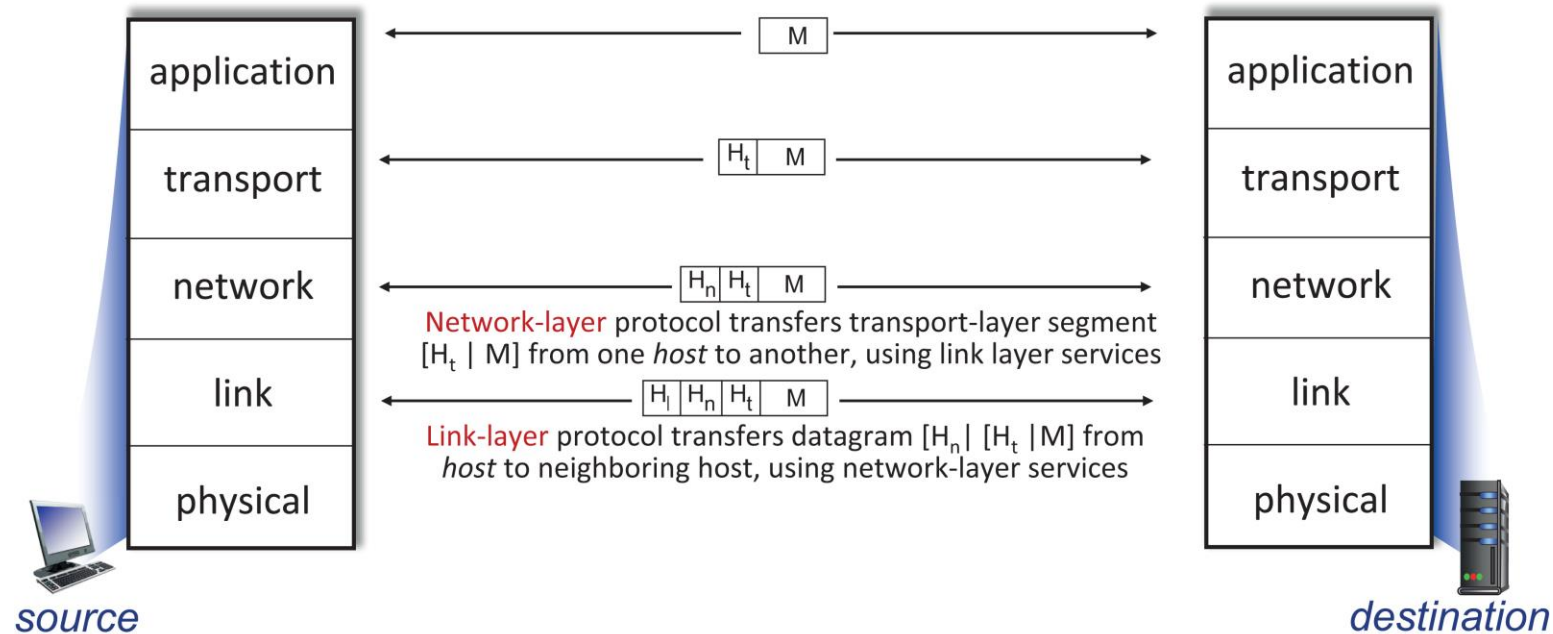
- transport-layer protocol **encapsulates** application-layer message, **M**, with **transport** layer-layer header **H_t** to create a transport-layer **segment**
- **H_t** used by transport layer protocol to implement its service

Services, Layering and Encapsulation (2 of 5)



- network-layer protocol **encapsulates** transport-layer segment $[H_t | M]$ with network layer-layer header H_n to create a network-layer **datagram**
- H_n used by network layer protocol to implement its service

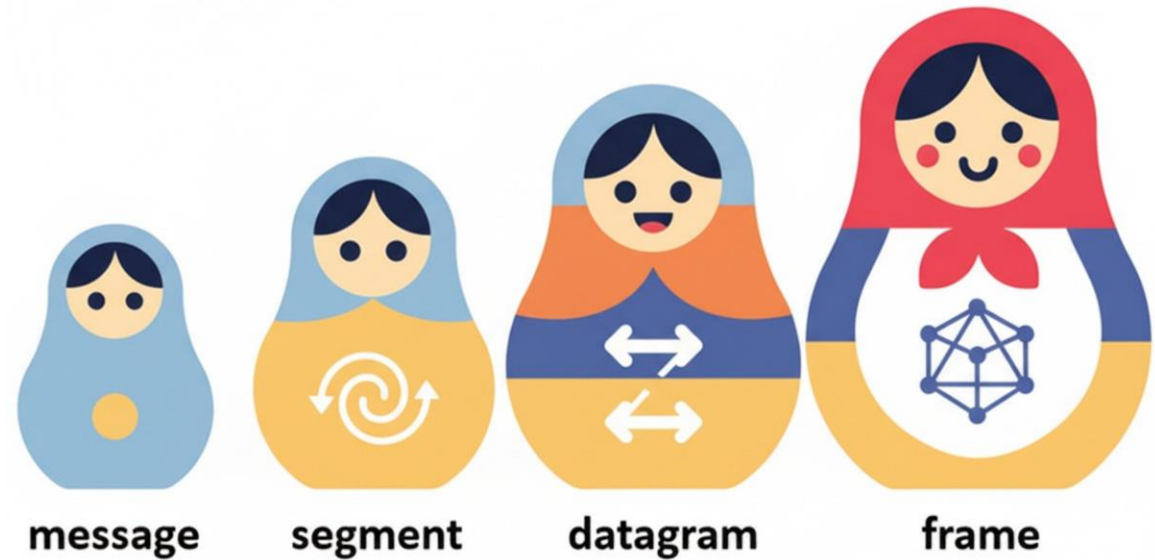
Services, Layering and Encapsulation (3 of 5)



- link-layer protocol **encapsulates** network datagram $[H_n | [H_t | M]]$ with link-layer header H_l to create a link-layer **frame**

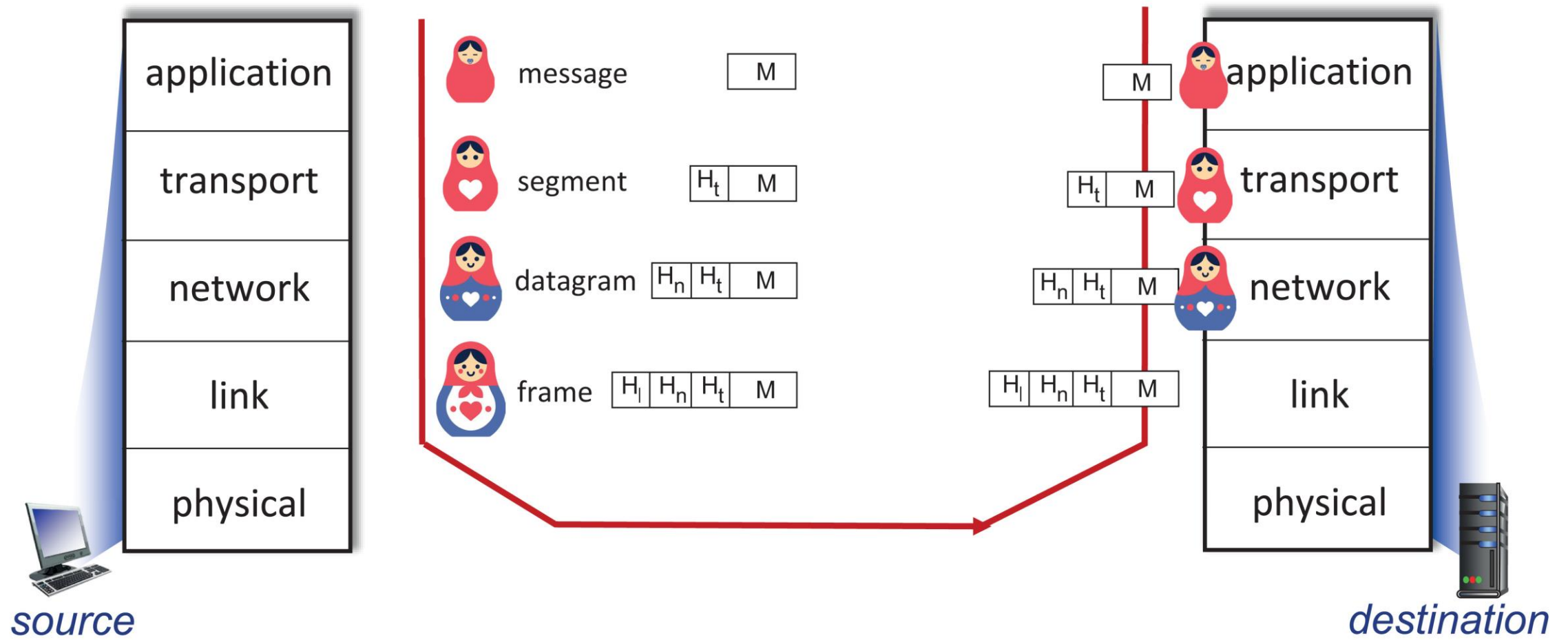
Encapsulation

Matryoshka dolls (stacking dolls)



Credit: <https://dribbble.com/shots/7182188-Babushka-Boi>

Services, Layering and Encapsulation (4 of 5)



Encapsulation: an End-End View

